Econometrics of Commodity and Asset Pricing

Lecture 1

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Agenda - Lecture 1

I. Discrete-Time Affine Processes

II. Pricing and Risk-Neutral Dynamics

Chapter I

DISCRETE TIME AFFINE PROCESSES

– Agenda –

I.1 – INFORMATION IN THE ECONOMY: THE "FACTORS"

- New information at date $t = 1, 2, \ldots, T$: n-dimensional "factor" w_t, or "state".

- Information available at t: $\underline{w_t} = (w_t, w_{t-1}, \ldots, w_1)$.

- w_t: random vector, observable, partially observable, or unobservable by the econometrician.

Example I.1

- *Option pricing:* $w_t = (y_t, z_t')'$ *with*
  y_t*: observable vector of (geometric) returns,*
  z_t*: regime, unobservable by the econometrician.*

- *Term structure of interest rates:*
  w_t*: unobservable factor or observable vector of yields or spreads, macroeconomic variables.*

I.2 – DYNAMIC MODELS

- Aim: modelling the dynamics of w_t (historical or risk-neutral, see Chapter II) in order to price (by no-arbitrage) the payoff of a given asset.
- Under the AAO and other standard assumptions (see Harrison and Kreps (1979) and Hansen and Richard (1987)):

- there exists a sequence of positive random variables $\mathcal{M}_{t,t+1} = \mathcal{M}_{t,t+1}(\underline{w_t})$, called stochastic discount factors (SDF), such that
- the date-t price $p_t[g(\underline{w_{t+h}})]$ (say) of the payoff $g(\underline{w_{t+h}})$ paid in $(t+h)$ is given by:

$$\begin{aligned} p_t[g(\underline{w_{t+h}})] &= \mathbb{E}[\mathcal{M}_{t,t+1} \dots \mathcal{M}_{t+h-1,t+h}\, g(\underline{w_{t+h}}) \mid \underline{w_t}], \\ &= \mathbb{E}[\mathcal{M}_{t,t+h}\, g(\underline{w_{t+h}}) \mid \underline{w_t}] \end{aligned} \tag{I.1}$$

- If any $\mathcal{M}_{t,t+1}$ (and, thus, $\mathcal{M}_{t,t+h}$) and $g(\underline{w_{t+h}})$ are exponential affine functions of $(w_t, \dots, w_{t+h})$, then the pricing formula (I.1) is a multi-horizon conditional Laplace transform.
- We want (I.1) to be explicit or quasi-explicit.

I.2 – DYNAMIC MODELS

☐ Parametric modelling:

- Choice of a family of conditional probability density functions $f(w_{t+1}|\underline{w_t}, \theta)$, $\theta \in \Theta$ (vector of unknown parameters).

- Equivalently, choice of . . .

  . . . a conditional Laplace transforms:

$$\varphi(u \mid \underline{w_t}, \theta) = \mathbb{E}_\theta[\exp(u' w_{t+1}) \mid \underline{w_t}], \quad u \in \mathbb{R}^n,$$

  . . . or a conditional log Laplace transforms:

$$\psi(u \mid \underline{w_t}, \theta) = \log\{\mathbb{E}_\theta[\exp(u' w_{t+1}) \mid \underline{w_t}]\}, \quad u \in \mathbb{R}^n.$$

Example I.2 (Conditionally Bernoulli process)

If w_{t+1} is valued in $\{0,1\}$, then there exists a function p such that w_{t+1} is conditionally Bernoulli, that is $w_{t+1}|\underline{w_t} \sim \mathcal{I}[p(\underline{w_t},\theta)]$, and:

$$\varphi_t(u) = \mathbb{E}\left[\exp(uw_{t+1})\,\middle|\underline{w_t}\right] = p_t\exp(u) + (1-p_t),$$

with $p_t = p(\underline{w_t},\theta)$.

Example I.3 (Conditionally Binomial process)

If $w_{t+1}|\underline{w_t} \in \mathcal{B}(\ell,p_t)$, valued in $\{1,\dots,\ell\}$, then:

$$\varphi(u|w_t) = [p_t\ \exp(u) + 1 - p_t]^{\ell}.$$

Example I.4 (Conditionally Poisson process)

A Poisson process is also valued in the set of integers, but contrary to the conditionally Bernoulli process, it is not upper bounded. More precisely, if $w_{t+1}|\underline{w_t} \sim \mathcal{P}(\lambda_t)$, then:

$$\begin{aligned}\varphi(u|w_t) &= \sum_{j=0}^{\infty}\frac{1}{j!}\exp(-\lambda_t)\lambda_t^j\exp(uj) = \exp(-\lambda_t)\exp[\lambda_t\exp(u)]\\ &= \exp\{\lambda_t[\exp(u)-1]\}.\end{aligned}$$

- The next example concerns Gaussian processes, which are among the most widely used processes in asset pricing.

Example I.5 (Conditionally normal (or Gaussian) process)

If $w_{t+1}|\underline{w_t} \sim \mathcal{N}\left(m(\underline{w_t},\theta), \Sigma(\underline{w_t},\theta)\right)$*, then:*

$$(u'w_{t+1})|\underline{w_t} \sim \mathcal{N}\left(u'm(\underline{w_t},\theta), u'\Sigma(\underline{w_t},\theta)u\right).$$

$$\Rightarrow \begin{cases} \varphi(u|\underline{w_t},\theta) &= \exp\left[\; u'm(\underline{w_t},\theta) + \frac{1}{2}u'\Sigma(\underline{w_t},\theta)u \;\right] \\ \psi(u|\underline{w_t},\theta) &= u'm(\underline{w_t},\theta) + \frac{1}{2}u'\Sigma(\underline{w_t},\theta)u. \end{cases}$$

- The equivalence between the conditional Laplace transform and the entire conditional distribution of w_t implies that the Laplace transform encodes all the conditional moments of w_t.
- If these moments exist, they can be easily extracted from the Laplace transform (see next section).

I.3 – SOME PROPERTIES OF THE LAPLACE TRANSFORM

- $\varphi(0|\underline{w_t},\theta) = 1$ [and $\psi(0|\underline{w_t},\theta) = 0$].
- Defined in a convex set E (containing 0).
- If the interior of E is non empty, all the (conditional) moments exist.
- Scalar case. Let us consider the conditional moment of order k. Given that:

$$\begin{aligned}
\frac{\partial^k}{\partial u^k}\mathbb{E}\left[\exp(uw_{t+1})\,\middle|\,\underline{w_t}\right] &= \mathbb{E}\left[\frac{\partial^k}{\partial u^k}\exp(uw_{t+1})\,\middle|\,\underline{w_t}\right] \\
&= \mathbb{E}\left[w_{t+1}^k\exp(uw_{t+1})\,\middle|\,\underline{w_t}\right],
\end{aligned}$$

we have:

$$\left[\frac{\partial^k\varphi(u|\underline{w_t},\theta)}{\partial u^k}\right]_{u=0} = \mathbb{E}_\theta[w_{t+1}^k|\underline{w_t}].$$

The conditional cumulant of order k is given by:

$$\left[\frac{\partial^k\psi(u|\underline{w_t},\theta)}{\partial u^k}\right]_{u=0} = \mathcal{C}_k(\underline{w_t},\theta).$$

We have, in particular:

$$\left\{\begin{array}{lcl}\mathcal{C}_1(\underline{w_t},\theta) & = & \mathbb{E}_\theta[w_{t+1}|\underline{w_t}] \\ \mathcal{C}_2(\underline{w_t},\theta) & = & \mathbb{V}ar_\theta[w_{t+1}|\underline{w_t}].\end{array}\right.$$

φ and ψ are referred to as the conditional moment-generating function and the conditional cumulant-generating function, respectively.

- Multivariate case:

$$\left[\frac{\partial\psi}{\partial u}(u|\underline{w_t},\theta)\right]_{u=0} = \mathbb{E}_\theta[w_{t+1}|\underline{w_t}]$$

$$\left[\frac{\partial^2\psi}{\partial u\partial u'}(u|\underline{w_t},\theta)\right]_{u=0} = \mathbb{V}ar_\theta[w_{t+1}|\underline{w_t}].$$

- Another property of the Laplace transform (useful in several pricing contexts). For $k \in \mathbb{N}$, and if this expectation exists, we have:

$$\mathbb{E}\left[(\nu_1' w_{t+1})^k \exp(\nu_2' w_{t+1})\,\middle|\underline{w_t}\right] = \left.\frac{\partial^k \varphi_t(s\nu_1+\nu_2)}{\partial s^k}\right|_{s=0}.$$

Indeed, we have $\left.\dfrac{\partial^k \exp[(s\nu_1+\nu_2)' w_{t+1}]}{\partial s^k}\right|_{s=0} = (\nu_1' w_{t+1})^k \exp(\nu_2' w_{t+1})$.

Example I.6 (Conditionally normal (or Gaussian) process (see Ex. I.5))

Scalar case:

- $\psi(u|\underline{w_t}, \theta) = um(\underline{w_t}, \theta) + \frac{1}{2}u^2\sigma^2(\underline{w_t}, \theta)$.
  - $\left[\dfrac{\partial \psi}{\partial u}(u|\underline{w_t}, \theta) \right]_{u=0} = m(\underline{w_t}, \theta)$.
  - $\left[\dfrac{\partial^2 \psi}{\partial u^2}(u|\underline{w_t}, \theta) \right]_{u=0} = \sigma^2(\underline{w_t}, \theta)$.

Multidimensional case:

- $\psi(u|\underline{w_t}, \theta) = u'm(\underline{w_t}, \theta) + \frac{1}{2}u'\Sigma(\underline{w_t}, \theta)u$.
  - $\left[\dfrac{\partial \psi}{\partial u}(u|\underline{w_t}, \theta) \right]_{u=0} = m(\underline{w_t}, \theta)$.
  - $\left[\dfrac{\partial^2 \psi}{\partial u \partial u'}(u|\underline{w_t}, \theta) \right]_{u=0} = \Sigma(\underline{w_t}, \theta)$.

- *In both cases, conditional cumulants of order > 2 are equal to 0.*

Conditional zero probability for non-negative processes.

- When certain components of w_{t+1} are non-negative, its cond. Laplace transform admits finite limits when the arguments associated with these components approach negative infinity.
- This property has been utilized in a term-structure context by Monfort, Pegoraro, Renne and Roussellet (2017, JOE) and Roussellet (2023, JOE) to calculate the conditional probabilities of future nominal short-term rates reaching the zero lower bound.

Lemma 1

Let w_t be univariate and nonnegative, and $\varphi_t(u) = \mathbb{E}_t[\exp(uw_{t+1})]$ is its (conditional) Laplace transform (defined for $u \le 0$). Then, we have:

$$\mathbb{P}_t(w_{t+1} = 0) = \lim_{u \to -\infty} \varphi_t(u).$$

Proof. We have:

$$\varphi_t(u) = \mathbb{P}_t(w_{t+1} = 0) + \int_{w_{t+1} > 0} \exp(uw_{t+1}) d\mathbb{P}_t(w_{t+1}) .$$

The Lebesgue theorem ensures that the last integral converges to zero when u goes to $-\infty$. ∎

Lemma 2

Let us consider the process $w_t = (w'_{1,t}, w_{2,t})'$ and assume that:

- $w_{1,t}$ *is valued in* $\mathbb{R}^d$ $(d \geq 1)$, *and* $w_{2,t}$ *is valued in* $\mathbb{R}^+ = [0, +\infty)$;
- $\mathbb{E}_t \left[\exp\left(u'_1 w_{1,t+1} + u_2 w_{2,t+1}\right)\right]$ *exists for a given* u_1 *and* $u_2 \leq 0$.

Then, we have:

$$\mathbb{E}_t \left[\exp(u'_1 w_{1,t+1})\, \mathbb{1}_{\{w_{2,t+1}=0\}}\right] = \lim_{u_2 \to -\infty} \mathbb{E}_t \left[\exp(u'_1 w_{1,t+1} + u_2\, w_{2,t+1})\right]. \tag{I.2}$$

Proof. We have that:

$$\lim_{u_2 \to -\infty} \mathbb{E}_t \left[\exp(u'_1 w_{1,t+1} + u_2\, w_{2,t+1})\right] = \mathbb{E}_t \left[\exp(u'_1 w_{1,t+1})\, \mathbb{1}_{\{w_{2,t+1}=0\}}\right] + \lim_{u_2 \to -\infty} \mathbb{E}_t \left[\exp(u'_1 w_{1,t+1} + u_2\, w_{2,t+1})\, \mathbb{1}_{\{w_{2,t+1}>0\}}\right].$$

For $w_{2,t+1} > 0$, $\exp(u'_1 w_{1,t+1} + u_2\, w_{2,t+1})\, \mathbb{1}_{\{w_{2,t+1}>0\}}$ converges to zero as $u_2 \to -\infty$, and it is bounded above by $\exp(u'_1 w_{1,t+1})$, which is integrable. By the Lebesgue dominated convergence theorem, the second term on the right-hand side therefore goes to zero as $u_2 \to -\infty$, which proves (I.2). If we take $u_1 = 0$, we have $\mathbb{P}\left[w_{2,t+1} = 0 \,\middle|\, \underline{w_t}\right] = \lim_{u_2 \to -\infty} \mathbb{E}\left[\exp(u_2 w_{2,t+1}) \,\middle|\, \underline{w_t}\right]$ (in line with previous lemma). ■

I.4 – AFFINE (OR Car) PROCESSES

a. Definition Affine process of order 1

- This section defines the affine processes of order one and provides several examples. In the following section, we will introduce the affine processes of order p and demonstrate that they admit a companion form that is affine of order one.
- The discrete-time affine processes have been introduced and studied under the name Compound Auto-Regressive (Car) processes by Darolles, Gourieroux and Jasiak (2006, JTSA).

Definition I.1 (Affine process of order 1)

A multivariate process w_{t+1} is Affine of order 1 (or Car(1)) if

$$\varphi_t(u) = \mathbb{E}_t[\exp(u' w_{t+1})] = \exp[a(u)' w_t + b(u)]$$

for some functions $a(.)$ and $b(.)$ (univariate if w_{t+1} and u are scalar).

- $a(.)$ and b$(.)$ may be deterministic functions of time.

Example I.7 (Univariate Gaussian $AR(1)$)

If $w_{t+1}|\underline{w_t} \sim \mathcal{N}(\nu + \rho w_t, \sigma^2)$, *then:*

$$\varphi_t(u) = \exp\left(u\rho w_t + u\nu + u^2\frac{\sigma^2}{2}\right) = \exp[a(u)\, w_t + b(u)],$$

$$\textit{with} \left\{ \begin{array}{rcl} a(u) & = & u\rho \\ b(u) & = & u\nu + u^2\dfrac{\sigma^2}{2}. \end{array} \right.$$

Example I.8 (Gaussian $VAR(1)$)

If $w_{t+1}|\underline{w_t} \sim \mathcal{N}(\mu + \Phi w_t, \Sigma)$, *then:*

$$\varphi_t(u) = \exp\left[\; u'(\mu + \Phi w_t) + \tfrac{1}{2}u'\Sigma u \;\right] = \exp[\, a(u)' w_t + b(u)\,],$$

$$\textit{with} \left\{ \begin{array}{rcl} a(u) & = & \Phi' u \\ b(u) & = & u'\mu + \frac{1}{2}u'\Sigma u = u'\mu + \frac{1}{2}(u \otimes u)' vec(\Sigma). \end{array} \right.$$

Example I.9 (Quadratic Gaussian process)

- *Consider $w_t = (x_t', vec(x_t x_t')')'$, where x_t is a n-dimensional vector following a Gaussian VAR(1), i.e.*

$$x_{t+1}|\underline{x_t} \sim \mathcal{N}(\mu + \Phi x_t, \Sigma),$$

then, the augmented state vector w_t is also affine. Let $u = (v, V)$ where $v \in \mathbb{R}^n$ and V is a $(n \times n)$ symmetric matrix such that $(I_n - 2\Sigma V)$ has strictly positive eigenvalues (i.e., the eigenvalues of ΣV, or $\Sigma^{1/2} V \Sigma^{1/2}$, are smaller than $1/2$), we have:

$$\begin{aligned}
\varphi_t(u) &= \mathbb{E}_t\left\{ \exp\left[(v', vec(V)') \times w_{t+1}\right]\right\} \\
&= \mathbb{E}_t[\exp(v' x_{t+1} + x_{t+1}' V x_{t+1})] \\
&= \exp\left\{a_1(v, V)' x_t + x_t' a_2(v, V)\, x_t + b(v, V)\right\},
\end{aligned}$$

where:

$$\begin{aligned}
a_1(u) &= \Phi' \left[(I_n - 2V\Sigma)^{-1}(v + 2V\mu)\right], \\
a_2(u) &= \Phi' V (I_n - 2\Sigma V)^{-1} \Phi, \\
b(u) &= v'(I_n - 2\Sigma V)^{-1}\left(\mu + \frac{1}{2}\Sigma v\right) + \\
&\qquad \mu' V (I_n - 2\Sigma V)^{-1} \mu - \frac{1}{2} \log \left| I_n - 2\Sigma V \right|.
\end{aligned}$$

The proof makes use of Lemmas 3 and 4 and of the properties of the *vec* operator.

Lemma 3

If $\mu \in \mathbb{R}^n$ and Q is a $(n \times n)$ matrix symmetric positive definite, then:

$$\int_{\mathbb{R}^n} \exp(-u'Qu + \mu'u)du = \frac{\pi^{n/2}}{(det\ Q)^{1/2}} \exp\left(\tfrac{1}{4}\mu' Q^{-1}\mu \right).$$

Proof. The LHS integral can be equivalently written as:

$$\int_{\mathbb{R}^n} \exp\left[-(u - \tfrac{1}{2}Q^{-1}\mu)' Q (u - \tfrac{1}{2}Q^{-1}\mu) \right] \exp\left(\tfrac{1}{4}\mu' Q^{-1}\mu \right) du$$

$$= \frac{\pi^{n/2}}{(det Q)^{1/2}} \exp\left(\tfrac{1}{4}\mu' Q^{-1}\mu \right),$$

which gives the result (the RHS) using the formula for the unit mass of a random variable $X \sim \mathcal{N}(0.5Q^{-1}\mu, (2Q)^{-1})$ with (obviously) pdf

$$f_X(x) = \frac{(det\ Q)^{1/2}}{\pi^{n/2}} \exp\left[-(x - \frac{1}{2}Q^{-1}\mu)' Q (x - \frac{1}{2}Q^{-1}\mu) \right].$$

■

Lemma 4

If $\varepsilon_{t+1} \sim \mathcal{N}(0, I_n)$, $\lambda \in \mathbb{R}^n$ and S is a $(n \times n)$ symmetric matrix with eigenvalues smaller than $1/2$, we have

$$\mathbb{E}_t \left(\exp[\lambda' \varepsilon_{t+1} + \varepsilon'_{t+1} S \varepsilon_{t+1}] \right) = \frac{1}{[\det(I_n - 2S)]^{1/2}} \exp \left[\frac{1}{2} \lambda' (I_n - 2S)^{-1} \lambda \right].$$

Proof. We have

$$\mathbb{E}_t \exp(\lambda' \varepsilon_{t+1} + \varepsilon'_{t+1} S \varepsilon_{t+1}) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \exp \left[-u' \left(\frac{1}{2} I_n - S \right) u + \lambda' u \right] du$$

From Lemma 3, if $u \in \mathbb{R}^n$, we know that

$$\int_{\mathbb{R}^n} \exp(-u' Q u + \mu' u) du = \frac{\pi^{n/2}}{(\det Q)^{1/2}} \exp \left(\frac{1}{4} \mu' Q^{-1} \mu \right).$$

Therefore, for $Q = \left(\frac{1}{2} I_n - S \right)$, we have:

$$\mathbb{E}_t \left(\exp[\lambda' \varepsilon_{t+1} + \varepsilon'_{t+1} S \varepsilon_{t+1}] \right) =$$

$$= \frac{1}{2^{n/2} \left[\det \left(\frac{1}{2} I_n - S \right) \right]^{1/2}} \exp \left[\frac{1}{4} \lambda' \left(\frac{1}{2} I_n - S \right)^{-1} \lambda \right]. \blacksquare$$

Example I.9 (Proof)

We want to calculate:

$$\mathbb{E}_t[\exp(v'x_{t+1} + vec(V)'vec(x_{t+1}x'_{t+1}))] = \mathbb{E}_t[\exp(v'x_{t+1} + x'_{t+1}V\,x_{t+1}))] \,.$$

Let consider the term in the expectation, and let us substitute x_{t+1} with $(\mu + \Phi x_t + \Sigma^{\frac{1}{2}}\varepsilon_{t+1})$. We obtain:

$$\begin{aligned} & \exp[v'x_{t+1} + x'_{t+1}V\,x_{t+1}] \\ = \quad & \exp[v'(\mu + \Phi x_t) + \mu' V \mu + 2\mu' V \Phi V + x'_t \Phi' V\,\Phi x_t] \times \\ & \quad \exp\left\{[v'\Sigma^{\frac{1}{2}} + 2(\mu + \Phi x_t)' V \Sigma^{\frac{1}{2}}]\varepsilon_{t+1} + \varepsilon'_{t+1}[\Sigma^{\frac{1}{2}\,\prime} V \Sigma^{\frac{1}{2}}]\varepsilon_{t+1}\right\} \,. \end{aligned}$$

Observe that the first term depends on the date-t information only, while we can apply Lemma 4 to the second term with $\lambda = \Sigma^{\frac{1}{2}\,\prime}(v + 2V'(\mu + \Phi x_t))$ and $S = \Sigma^{\frac{1}{2}\,\prime} V \Sigma^{\frac{1}{2}}$. Some algebraic computation leads to the following result:

Example I.9 (Proof)

$$\mathbb{E}_t\left[\exp\left\{[v'\Sigma^{\frac{1}{2}} + 2(\mu+\Phi x_t)'V\Sigma^{\frac{1}{2}}]\varepsilon_{t+1} + \varepsilon'_{t+1}[\Sigma^{\frac{1}{2}\prime}V\Sigma^{\frac{1}{2}}]\varepsilon_{t+1}\right\}\right] =$$

$$\exp\left\{-\frac{1}{2}\log\det\left[I_n - 2(\Sigma^{\frac{1}{2}\prime}V\Sigma^{\frac{1}{2}})\right] + \frac{1}{2}v'\Sigma^{\frac{1}{2}}\left[I_n - 2(\Sigma^{\frac{1}{2}\prime}V\Sigma^{\frac{1}{2}})\right]^{-1}\Sigma^{\frac{1}{2}\prime}v\right.$$

$$+2v'\Sigma^{\frac{1}{2}}\left[I_n - 2(\Sigma^{\frac{1}{2}\prime}V\Sigma^{\frac{1}{2}})\right]^{-1}\Sigma^{\frac{1}{2}\prime}V\mu + 2\mu'V\Sigma^{\frac{1}{2}}\left[I_n - 2(\Sigma^{\frac{1}{2}\prime}V\Sigma^{\frac{1}{2}})\right]^{-1}\Sigma^{\frac{1}{2}\prime}V\mu$$

$$+\left[2v'\Sigma^{\frac{1}{2}}\left[I_n - 2(\Sigma^{\frac{1}{2}\prime}V\Sigma^{\frac{1}{2}})\right]^{-1}\Sigma^{\frac{1}{2}\prime}V\Phi + 4\mu'V\Sigma^{\frac{1}{2}}\left[I_n - 2(\Sigma^{\frac{1}{2}\prime}V\Sigma^{\frac{1}{2}})\right]^{-1}\Sigma^{\frac{1}{2}\prime}V\Phi\right]x_t$$

$$\left.+x_t'\left[2\Phi'V\Sigma^{\frac{1}{2}}\left[I_n - 2(\Sigma^{\frac{1}{2}\prime}V\Sigma^{\frac{1}{2}})\right]^{-1}\Sigma^{\frac{1}{2}\prime}V\Phi\right]x_t\right\}.$$

Putting together the first and second term we obtain:

$$\varphi_t(v,V) = \exp\left[a_1(v,V)'x_t + x_t'a_2(v,V)x_t + b(v,V)\right],$$

where:

Example I.9 (Proof)

$$
\begin{aligned}
a_1(v,V) &= \Phi' \Big\{ v + 2V\mu + 2V\Sigma^{\frac{1}{2}} \left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}}) \right]^{-1} \Sigma^{\frac{1}{2}\prime} v \\
&\qquad + 4V\Sigma^{\frac{1}{2}} \left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}}) \right]^{-1} \Sigma^{\frac{1}{2}\prime} V\mu \Big\} \\
a_2(v,V) &= \Phi' \left[V + 2V\Sigma^{\frac{1}{2}} \left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}}) \right]^{-1} \Sigma^{\frac{1}{2}\prime} V \right] \Phi \\
b(v,V) &= v'\mu + \mu' V\mu - \frac{1}{2} \log\det \left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}}) \right] \\
&\quad + \frac{1}{2} v'\Sigma^{\frac{1}{2}} \left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}}) \right]^{-1} \Sigma^{\frac{1}{2}\prime} v \\
&\quad + 2v'\Sigma^{\frac{1}{2}} \left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}}) \right]^{-1} \Sigma^{\frac{1}{2}\prime} V\mu \\
&\quad + 2\mu' V\Sigma^{\frac{1}{2}} \left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}}) \right]^{-1} \Sigma^{\frac{1}{2}\prime} V\mu .
\end{aligned}
$$

Example I.9 (Proof)

Then, given that $\left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}})\right] = \Sigma^{\frac{1}{2}\prime}\left[\Sigma^{-1} - 2V\right]\Sigma^{\frac{1}{2}}$, *we have:*

$$\left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}})\right]^{-1} = \Sigma^{-\frac{1}{2}}\left[\Sigma^{-1} - 2V\right]^{-1}(\Sigma^{-\frac{1}{2}})',$$

from which we obtain:

$$\left[\Sigma^{-1} - 2V\right]^{-1} = \Sigma^{\frac{1}{2}}\left[I_n - 2(\Sigma^{\frac{1}{2}\prime} V \Sigma^{\frac{1}{2}})\right]^{-1}\Sigma^{\frac{1}{2}\prime}.$$

Observe also that:

$$\left[\Sigma^{-1} - 2V\right]^{-1} = [I_n - 2\Sigma V]^{-1}\Sigma.$$

Example I.9 (Proof)

We can thus simplify the previous expressions and obtain:

$$
\begin{aligned}
a_1(v,V) &= \Phi'(v+2V\mu) + 2\Phi' V \left[I_n - 2\Sigma V\right]^{-1} \Sigma (v+2V\mu) \\
&= \Phi' \left\{ 2\, V\, \left[I_n - 2\Sigma V\right]^{-1} \Sigma + I_n \right\} (v+2V\mu) \\
&= \Phi' \left[I_n - 2\Sigma V\right]^{-1} (v+2V\mu)\,;
\end{aligned}
$$

$$
\begin{aligned}
a_2(v,V) &= \Phi' \left\{ V + 2\, V\, \left[I_n - 2\Sigma V\right]^{-1} \Sigma\, V \right\} \\
&= \Phi' V \left[I_n - 2\Sigma V\right]^{-1} \Phi\,;
\end{aligned}
$$

$$
\begin{aligned}
b(v,V) &= \mu' V \left[\mu + 2\left(I_n - 2\Sigma V\right)^{-1} \Sigma\, V\, \mu\right] \\
&\quad + v' \left\{ \mu + \left[I_n - 2\Sigma V\right]^{-1} \Sigma \left(\frac{1}{2}\, v + 2\, V\, \mu \right) \right\} \\
&\qquad - \frac{1}{2} \log\det\left[I_n - 2\Sigma V\right] \\
&= \mu' V \left[I_n - 2\Sigma V\right]^{-1} \mu + v' \left[I_n - 2\Sigma V\right]^{-1} \left(\mu + \frac{1}{2} \Sigma v \right) \\
&\qquad - \frac{1}{2} \log\det\left[I_n - 2\Sigma V\right]\,. \qquad \blacksquare
\end{aligned}
$$

Example I.10 (The Non-centered Gamma Distribution)

Let us assume that Y is a Non-centered Gamma random variable with parameters $\nu > 0$, $\beta > 0$ and $\mu > 0$, i.e. $Y \sim \widetilde{\gamma}(\nu, \beta, \mu)$. We know that this is equivalent to the existence of a random variable $Z \sim \mathcal{P}(\beta)$ such that:

$$\left\{ \begin{array}{l} \dfrac{Y}{\mu} \,|\, Z \sim \gamma(\nu + Z)\,, \;\; \nu > 0\,, \\ \\ Z \sim \mathcal{P}(\beta)\,, \;\; \beta > 0\,, \mu > 0\,, \end{array} \right. \iff \left\{ \begin{array}{l} Y \,|\, Z \sim \gamma(\nu + Z, \mu)\,, \;\; \nu > 0\,, \\ \\ Z \sim \mathcal{P}(\beta)\,, \;\; \beta > 0\,, \mu > 0\,, \end{array} \right.$$

where $\gamma(\nu, \mu)$ denotes the (centered) Gamma distribution of shape (d.o.f.) parameter ν and of scale parameter μ, and β is the non-centrality parameter. We have that:

- $\mathbb{E}[Y] = \nu\,\mu + \beta\,\mu$ *and* $\mathbb{V}[Y] = \nu\,\mu^2 + 2\mu^2\beta$ *(**mean** and **variance**);*
- $\mathbb{E}[\exp(uY)] = \exp\left[-\nu\log(1 - u\,\mu) + \beta\dfrac{u\,\mu}{1 - u\,\mu}\right]$*, for $u < 1/\mu$ (**Laplace transform**).*

Example I.10 (Proof)

i) *Given that* $Y \mid Z \sim \gamma(\nu + Z, \mu)$ *and that* $\mathbb{E}(Y) = \mathbb{E}[\mathbb{E}(Y \mid Z)]$ *we easily find that:*

$$\mathbb{E}(Y) = \mathbb{E}[\mu(\nu + Z)] = \mu\,\nu + \mu\,\mathbb{E}(Z) = \mu\,\nu + \mu\,\beta.$$

ii) *Given that* $\mathbb{V}(Y) = \mathbb{E}[\mathbb{V}(Y \mid Z)] + \mathbb{V}[\mathbb{E}(Y \mid Z)]$, *again we easily find:*

$$\begin{aligned}\mathbb{V}(Y) &= \mathbb{E}[\mathbb{V}(Y \mid Z)] + \mathbb{V}[\mathbb{E}(Y \mid Z)] \\ &= \mathbb{E}[\mu^2(\nu + Z)] + \mathbb{V}[\mu(\nu + Z)] \\ &= \nu\,\mu^2 + 2\mu^2\beta\,.\end{aligned}$$

iii) *With regard to the Laplace transform, we have:*

$$\begin{aligned}\varphi(u) &= \mathbb{E}[\exp(u\,Y)] = \mathbb{E}_Z\left[\mathbb{E}_{Y\mid Z}(\exp(uY) \mid Z)\right] \\ &= \mathbb{E}_Z\left[\left(\frac{1}{1-\mu u}\right)^{\nu+Z}\right] = \left(\frac{1}{1-\mu u}\right)^{\nu} \mathbb{E}_Z\left[\left(\frac{1}{1-\mu u}\right)^{Z}\right] \\ &= \exp[-\nu\log(1-\mu u)]\,E_Z[\exp(-Z\,\log(1-\mu\,u))] \\ &= \exp[-\nu\log(1-\mu u)]\,\exp\left[\beta\,\frac{u\mu}{1-u\mu}\right] \\ &= \exp\left[-\nu\log(1-\mu u) + \beta\,\frac{u\mu}{1-u\mu}\right].\end{aligned}$$

Example I.11 (Autoregressive Gamma process, ARG(1))

The Autoregressive Gamma of order one [ARG(1)] process (w_t) (say) is the exact discrete-time equivalent of the square-root process introduced in the continuous-time term structure literature by Cox, Ingersoll and Ross (1985). This process can be defined as:

$$\begin{cases} \dfrac{w_{t+1}}{\mu} | z_{t+1} \sim \gamma(\nu + z_{t+1}), \;\; \nu > 0, \\ \\ z_{t+1} | w_t \sim \mathcal{P}(\rho w_t/\mu), \;\; \rho > 0, \mu > 0, \end{cases}$$

where $\gamma(\nu, \mu)$ is the (centered) Gamma distribution, $\mathcal{P}(\lambda)$ a Poisson distr., $\mu > 0$ is the scale parameter, $\nu > 0$ is the d.o.f. parameter, $\rho > 0$ is the AR parameter, and z_t is the mixing variable. Alternatively $z_t \sim \mathcal{P}(\beta w_t)$, with $\rho = \beta\mu$.

We have: $\varphi_t(u) = exp\left[\;\dfrac{\rho u}{1 - u\mu} w_t - \nu \log(1 - u\mu) \;\right], u < \dfrac{1}{\mu}.$

That is: $\varphi_t(u) = \exp[a(u)' w_t + b(u)]$ *with (see next slide)*

$$\begin{cases} a(u) &= \dfrac{\rho u}{1 - u\mu} \\ b(u) &= -\nu \log(1 - u\mu). \end{cases}$$

▸ *Web-interface illustration ("ARG" panel)*

Example I.11 (Autoregressive Gamma process, ARG(1) (Laplace Transform: proof))

Given $\underline{w_t}$, we have $z_{t+1} \sim \mathcal{P}\left(\frac{\rho w_t}{\mu}\right)$. We have:

$$
\begin{aligned}
\mathbb{E}[\exp(uw_{t+1})|\underline{w_t}] &= \mathbb{E}\left\{\mathbb{E}\left[\exp\left(u\mu\frac{w_{t+1}}{\mu}\right)|\underline{w_t}, \underline{z_{t+1}}\right]\underline{w_t}\right\} \\
&= \mathbb{E}[(1-u\mu)^{-(\nu+z_{t+1})}|\underline{w_t}] \\
&= (1-u\mu)^{-\nu}\mathbb{E}\{\exp[-z_{t+1}\log(1-u\mu)]|\underline{w_t}\} \\
&= (1-u\mu)^{-\nu}\exp\left\{\frac{\rho w_t}{\mu}[\exp(-\log(1-u\mu)] - \frac{\rho w_t}{\mu}\right\} \\
&= \exp\left[\frac{\rho u w_t}{1-u\mu} - \nu\log(1-u\mu)\right],
\end{aligned}
$$

using the fact that the L.T. of $\gamma(\nu)$ is $(1-u)^{-\nu}$ and that the L.T. of $\mathcal{P}(\lambda)$ is $\exp[\lambda(exp(u)-1)]$.

Example I.11 (Autoregressive Gamma process, ARG(1) (cont'd))

- *The conditional probability density function* $f(w_{t+1} \mid w_t; \mu, \nu, \rho)$ *(say) of the ARG(1) process is the following mixture of Gamma densities with Poisson weights:*

$$f(w_{t+1} \mid w_t; \mu, \nu, \rho) = \sum_{k=0}^{+\infty} \left[\frac{1}{\mu} \frac{e^{-\frac{w_{t+1}}{\mu}} \left(\frac{w_{t+1}}{\mu} \right)^{\nu+k-1}}{\Gamma(\nu+k)} \times \frac{\left(\frac{\rho w_t}{\mu}\right)^k}{k!} e^{-\frac{\rho w_t}{\mu}} \right],$$

where $\rho > 0\,, \mu > 0\,, \nu > 0\,,$

- (w_t) *is a positive process and it is (second-order) stationary if and only if* $0 < \rho < 1$. *We have:*

$$\begin{cases} \mathbb{E}(w_{t+1} | \underline{w_t}) & = & \nu\mu + \rho w_t \\ \mathbb{V}ar(w_{t+1} | \underline{w_t}) & = & \nu\mu^2 + 2\mu\rho w_t. \end{cases}$$

- *The positive process* (w_t) *has the following representation:*

$$w_{t+1} = \nu\mu + \rho w_t + \varepsilon_{t+1},$$

where ε_{t+1} *is a martingale difference* $\Rightarrow$ w_{t+1} *is a semi-strong* $AR(1)$.

Example I.11 (Autoregressive Gamma process, ARG(1) (cont'd))

- *More precisely, (ε_t) is a conditionally heteroskedastic martingale difference, whose conditional variance is*

$$\mathbb{V}ar(\varepsilon_{t+1} \,|\, \varepsilon_t) = \mathbb{V}ar(w_{t+1} \,|\, w_t) = \nu\mu^2 + 2\mu\rho w_t.$$

- *Under stationarity, the process (ε_t) has finite unconditional variance given by:*

$$\mathbb{V}ar(\varepsilon_t) = \nu\mu^2 + 2\nu\mu^2 \frac{\rho}{1-\rho}$$

- *The unconditional mean and variance of (w_t) are respectively given by*

$$\begin{aligned} \mathbb{E}(w_t) &= \frac{\nu\mu}{1-\rho} \\ \mathbb{V}ar(w_t) &= \frac{\nu\mu^2}{(1-\rho)^2}, \end{aligned}$$

- *See Gourieroux and Jasiak (2006) for further details.*

Example I.11 (Autoregressive Gamma-Zero process, ARG(1) (cont'd))

- *Let us consider now the Gamma-Zero distribution $\gamma_0(\lambda, \mu)$. We construct this new distribution in two steps (introducing a mixing variable Z):*

  1. $Z \sim \mathcal{P}(\lambda) \Longrightarrow Z(\omega) \in \{\mathbf{0}, 1, 2, \ldots\}$ *and* $\mathbb{P}(Z = 0) = \exp(-\lambda)$.
  2. *We define $W|Z \sim \gamma_Z(\mu)$, which implies:*
     - *If $Z = 0$, W is a dirac point mass at 0.*
     - *If $Z \neq 0$, W is gamma-distributed (continuous on $\mathbb{R}^+$).*

Definition

The non-negative r.v. $W \sim \gamma_0(\lambda, \mu)$, $\lambda > 0$ and $\mu > 0$, if

$$\begin{aligned} & W \,|\, Z \sim \gamma_Z(\mu) \quad \text{with} \quad Z \sim \mathcal{P}(\lambda) \\ \Rightarrow \qquad & \mathbb{P}(W = 0) = \mathbb{P}(Z = 0) = \exp(-\lambda)\,. \end{aligned}$$

- *In other words, $W \sim \gamma_0(\lambda, \mu)$ if its (complicated) p.d.f. is:*

$$f_W(w\,;\,\lambda, \mu) = \sum_{z=1}^{+\infty} \left[\frac{\exp(-w/\mu)\, w^{z-1}}{(z-1)!\,\mu^z} \times \frac{\exp(-\lambda)\lambda^z}{z\,!} \right] \mathbb{1}_{\{w>0\}} + \exp(-\lambda)\mathbb{1}_{\{w=0\}}$$

Example I.11 (Autoregressive Gamma-Zero process, ARG(1) (cont'd))

- *However, it is characterized by a simple Laplace transform (LT):*

$$\varphi_W(u\,;\,\lambda,\mu) := \mathbb{E}\left[\exp(uW)\right] = \exp\left[\lambda\frac{u\mu}{(1-u\mu)}\right] \quad \text{for} \quad u < \frac{1}{\mu}\,.$$

$\Longrightarrow$ *which is exponential-affine in* λ.

Definition

(w_t) *is a* $ARG_0(\alpha,\beta,\mu)$ *if* $(w_{t+1}|\underline{w_t})$ *is Gamma-Zero distributed:*

$$(w_{t+1}|\underline{w_t}) \sim \gamma_0(\alpha+\beta w_t,\mu) \quad \text{for} \quad \alpha \geq 0,\ \mu > 0,\ \beta > 0\,.$$

That is:

$$\frac{w_{t+1}}{\mu}\,|\,z_{t+1} \sim \gamma(z_{t+1}), \quad z_{t+1}\,|\,w_t \sim \mathcal{P}(\alpha+\beta w_t) \tag{I.3}$$

Again, simple conditional LT, exponential-affine in w_t*:*

$$\begin{aligned}\varphi_{w,t}(u\,;\,\alpha,\beta,\mu) &:= \mathbb{E}_t\left[\exp(uw_{t+1})\right] \\ &= \exp\left[\frac{u\mu}{1-u\mu}(\alpha+\beta\,w_t)\right], \quad \text{for} \quad u < \frac{1}{\mu}\,.\end{aligned}$$

Example I.11 (Autoregressive Gamma-Zero process, ARG(1) (cont'd))

- *The $ARG_0(\alpha, \beta, \mu)$ process is able to stay at zero with probability:*

$$\mathbb{P}(w_{t+1} = 0 | w_t = 0) = \exp(-\alpha) \neq 0.$$

given that

$$\mathbb{P}(w_{t+1} = 0 | w_t) = \exp(-\alpha - \beta w_t).$$

- *Zero is an absorbing state if $\alpha = 0$.*
- *The conditional mean and conditional variance of w_{t+1} given its past are respectively given by:*

$$\mathbb{E}_t(w_{t+1}) = \alpha\mu + \rho w_t \, , \ \mathbb{V}ar_t(w_{t+1}) = 2\mu^2\alpha + 2\mu\rho w_t = 2\mu\,\mathbb{E}_t(w_{t+1}) \, ,$$

with $\rho = \beta\,\mu$.

- *(w_t) has the following semi-strong AR(1) representation:*

$$w_{t+1} = \alpha\mu + \rho w_t + \varepsilon_{t+1} \, ,$$

where (ε_t) is a conditionally heteroskedastic martingale difference, whose conditional variance is $\mathbb{V}(\varepsilon_{t+1} \,|\, \underline{\varepsilon_t}) = 2\mu^2\alpha + 2\mu\rho w_t$.

Example I.11 (Autoregressive Gamma-Zero process, ARG(1) (cont'd))

- *(w_t) is stationary if and only if $\rho < 1$ and, in this case, its unconditional mean and variance are respectively given by:*

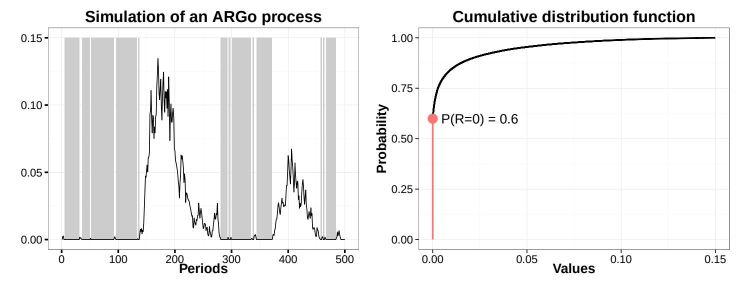
$$\mathbb{E}(w_t) = \frac{\alpha\mu}{1-\rho} \qquad \textit{and} \qquad \mathbb{V}ar(w_t) = \frac{2\alpha\mu^2}{(1-\rho)(1-\rho^2)}.$$

- *General case: $w_{t+1} \,|\, w_t \sim ARG_\nu(\alpha, \beta, \mu)$.*
  - $\frac{w_{t+1}}{\mu} \,|\, z_{t+1} \sim \gamma(\nu + z_{t+1})$ *with* $z_{t+1} \,|\, w_t \sim \mathcal{P}(\alpha + \beta\, w_t)$
  - $\nu = 0 \Longrightarrow ARG_0$ *process.*
  - $\nu > 0$, $\alpha = 0 \Longrightarrow$ *ARG process of Gourieroux & Jasiak (2006).*

	$\nu = 0$	$\nu > 0$
Positivity	*Yes*	*Yes*
Affine	*Yes*	*Yes*
Zero point mass	*Yes*	*No*

(Monfort, Pegoraro, Renne and Roussellet, 2017).

Example I.11 (Autoregressive gamma process, ARG(1) (cont'd))



[$ARG_0(1)$ simulation from Monfort, Pegoraro, Renne and Roussellet, 2017]

Example I.12 (Compound Poisson process)

$$\frac{w_{t+1}}{\gamma} = z_{t+1} + \varepsilon_{t+1},$$

where:

- z_{t+1} *and* ε_{t+1} *conditionally independent,*
- $z_{t+1} \sim \mathcal{B}in\left(\frac{w_t}{\gamma}, \pi \right)$,
- $\varepsilon_{t+1} \sim \mathcal{P}(\lambda)$,
- $\gamma > 0,\ 0 < \pi < 1,\ \lambda > 0$.

This process is valued in $\{j\gamma, j \in \mathbb{N}, \gamma \in \mathbb{R}^+\}$ *and we have:*

$$\varphi_t(u) = exp\left\{ \frac{w_t}{\gamma} \log[\pi\, exp(u\gamma) + 1 - \pi] - \lambda[1 - exp(u\gamma)] \right\},$$

with

$$\left\{ \begin{array}{lcl} a(u) & = & \frac{1}{\gamma} \log[\pi\ \exp(u\gamma) + 1 - \pi], \\ b(u) & = & -\lambda[1 - exp(u\gamma)]. \end{array} \right.$$

- *We have:* $w_{t+1} = \pi w_t + \lambda\gamma + \eta_{t+1}$, *where* η_{t+1} *is a martingale difference.*

Web-interface illustration ("Compound Poisson" panel)

I.4 – AFFINE (OR Car) PROCESSES

b. Definition Affine process of order p

Definition I.2 (Affine process of order p)

A multivariate process w_{t+1} is affine of order p [or $Car(p)$] if

$$\varphi_t(u) = \mathbb{E}_t[\exp(u'w_{t+1})] = \exp[a_1(u)'w_t + \dots + a_p(u)'w_{t+1-p} + b(u)].$$

Remark I.1

If w_t is $Car(p)$, then $W_t = [w_t', w_{t-1}', \dots, w_{t-p+1}']'$ is $Car(1)$.

$\Rightarrow$ *Without loss of generality we can assume $p = 1$.*

Proof. We have:

$$\begin{aligned}
\mathbb{E}_t[\exp(u'W_{t+1})] &= \mathbb{E}_t[\exp(u_1'w_{t+1} + u_2'w_t + \dots + u_p'w_{t-p+2})] \\
&= \exp(u_2'w_t + \dots + u_p'w_{t-p+2})\mathbb{E}_t[\exp(u_1'w_{t+1})] \\
&= \exp(u_2'w_t + \dots + u_p'w_{t-p+2} + a_1'(u_1)w_t + \dots + a_p'(u_1)w_{t+1-p} + b(u_1)) \\
&= \exp[A(u)'W_t + B(u)],
\end{aligned}$$

with $A(u)' = [u_2' + a_1'(u_1), \dots, u_p' + a_{p-1}'(u_1), a_p'(u_1)]$ and $B(u) = b(u_1)$. ■

An important particular case

Definition I.3 (Univariate Index Affine of order p)

Let $\exp[a(u)w + b(u)]$ be the L.T. of a univariate Affine process of order 1, the process w_{t+1} is an Index-Affine process of order p if:

$$\varphi_t(u) = \mathbb{E}_t[\exp(uw_{t+1})] = \exp[a(u)(\beta_1 w_t + \cdots + \beta_p w_{t+1-p}) + b(u)].$$

Example I.13 (Gaussian $AR(p)$ process / extends Ex. I.7)

$$\begin{aligned}
\varphi_t(u) &= \exp\left[u\rho(\beta_1 w_t + \cdots + \beta_p w_{t+1-p}) + u\nu + u^2\frac{\sigma^2}{2}\right] \\
w_{t+1} &= \nu + \varphi_1 w_t + \cdots + \varphi_p w_{t+1-p} + \sigma\varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim i.i.d.\mathcal{N}(0,1),
\end{aligned}$$

with $\varphi_i = \rho\beta_i$.

Example I.14 (ARG(p) process (positive) / extends Ex. I.11)

$$\begin{aligned}
\varphi_t(u) &= \exp\left[\frac{u\mu}{1-u\mu}(\beta_1 w_t + \cdots + \beta_p w_{t+1-p}) - \nu\log(1-u\mu)\right], \\
w_{t+1} &= \nu\mu + \varphi_1 w_t + \cdots + \varphi_p w_{t+1-p} + \varepsilon_{t+1},
\end{aligned}$$

with $\varphi_i = \mu\beta_i$ and where ε_{t+1} is a martingale difference.

I.4 – AFFINE (OR Car) PROCESSES

c. Homogeneous Markov Chains

- Notations
  - e_j denotes the j^{th} column of Id_J, $j = 1, \dots, J$.
  - $\pi(e_i, e_j) = \mathbb{P}(z_{t+1} = e_j \,|\, z_t = e_i)$.
- With these notations:

$$\mathbb{E}[\exp(v' z_{t+1}) \,|\, z_t = e_i, \underline{z_{t-1}}] = \sum_{j=1}^{J} \exp(v' e_j) \pi(e_i, e_j),$$

- That is:

$$\varphi_t(v) = \exp[a_z(v, \pi)' z_t],$$

with

$$a_z(v, \pi) = \begin{bmatrix} \log\left(\sum_{j=1}^{J} \exp(v' e_j) \pi(e_1, e_j) \right) \\ \vdots \\ \log\left(\sum_{j=1}^{J} \exp(v' e_j) \pi(e_J, e_j) \right) \end{bmatrix}.$$

▸ Web-interface illustration ("Markov-Switching" panel)

I.4 – AFFINE (OR Car) PROCESSES

c. Homogeneous Markov Chains

- **Proof:**

$$
\begin{aligned}
\varphi(v) &= \mathbb{E}[\exp(v' z_{t+1}) | \, z_t = e_i] \\
&= \prod_{i=1}^{J} \left[\sum_{j=1}^{J} \exp(v' e_j) \pi(e_i, e_j) \right]^{\mathbb{1}_{\{z_t = e_i\}}} \\
&= \exp \left[\sum_{i=1}^{J} \mathbb{1}_{\{z_t = e_i\}} \log \left(\sum_{j=1}^{J} \exp(v' e_j) \pi(e_i, e_j) \right) \right] \\
&= \exp(a_z(v, \pi)' z_t) \, ,
\end{aligned}
$$

- Observe that, if the Markov Chain is non-homogeneous, that is if the date-t conditioning information depends on a stochastic process (y_t):

$$\mathbb{P}(z_{t+1} = e_j \, | \, z_t = e_i, y_t) = \pi_t(e_i, e_j)$$

then (z_t) is not an affine process. Indeed, $a_z(v, \pi_t)$ will be function of y_t.

I.4 – AFFINE (OR Car) PROCESSES

d. Switching regimes univariate Car(1) processes

- The construction is based on the following steps:

a) Let us consider the previously introduced J-states homogeneous Markov Chain (z_t). We have seen that (z_t) is an affine (Car(1)) process.

b) Let us also consider a univariate Car(1) process with a conditional Laplace transform given by $\exp\left[a(u)\,w_t + b(u)\right]$, and let us assume that $b(u)$ can be written:

$$
\begin{aligned}
b(u) &= \tilde{b}(u)'\lambda \quad \text{where} \\
\tilde{b}(u) &= (b_1(u), \ldots, b_m(u))' \text{ and } \lambda = (\lambda_1, \ldots, \lambda_m)'\,.
\end{aligned}
$$

c) Let us assume that the parameters λ_i are stochastic and linear functions of z_{t+1}:

$$
\begin{aligned}
&\lambda_i(z_{t+1}) = \lambda_i' z_{t+1} \ \forall \ i \in \{1, \ldots, m\}\,, \\
&\lambda_i \in \mathbb{R}^J\,.
\end{aligned}
$$

I.4 – AFFINE (OR Car) PROCESSES

d. Switching regimes univariate Car(1) processes

- The conditional distribution of w_{t+1}, given $\underline{w_t}$ and $\underline{z_{t+1}}$, has a Laplace transform given by:

$$\mathbb{E}[\exp(uw_{t+1}) | \, \underline{w_t} \, , \underline{z_{t+1}}] = \exp\left[a(u)\, w_t + \tilde{b}(u)'\Lambda z_{t+1}\right] \; ,$$

$$\Lambda = \begin{bmatrix} \lambda_1' \\ \vdots \\ \lambda_m' \end{bmatrix} \text{ is a } (m, J)\text{-matrix.} \tag{I.4}$$

- The conditional Laplace transform of $(w_{t+1}, z_{t+1}')'$ given $\underline{w_t} \, , \underline{z_t}$ has the following form:

$$\begin{aligned} \varphi_t(u, v) &= \mathbb{E}\left[\exp(uw_{t+1} + v'z_{t+1}) \,|\, \underline{z_t}, \, \underline{w_t}\right] \\ &= \exp\left\{a(u)\, w_t + a_z(v + \Lambda'\tilde{b}(u), \pi)'z_t\right\} \; , \end{aligned}$$

and thus $(w_{t+1}, z_{t+1}')'$ is a Car(1) process.

- **Proof:**

$$
\begin{aligned}
\varphi_t(u,v) &= \mathbb{E}[\exp(uw_{t+1} + v'z_{t+1}) \mid \underline{w_t}\,,\underline{z_t}] \\
&= \mathbb{E}\left\{\mathbb{E}[\exp(uw_{t+1} + v'z_{t+1}) \mid \underline{w_t}\,,\underline{z_{t+1}}] \mid \underline{w_t}\,,\underline{z_t}\right\} \\
&= \mathbb{E}\left\{\exp(v'z_{t+1})\,\mathbb{E}[\exp(uw_{t+1}) \mid \underline{w_t}\,,\underline{z_{t+1}}] \mid \underline{w_t}\,,\underline{z_t}\right\} \\
&= \mathbb{E}[\exp(v'z_{t+1} + a(u)\,w_t + \tilde{b}(u)'\Lambda z_{t+1}) \mid \underline{w_t}\,,\underline{z_t}] \\
&= \exp[a(u)\,w_t]\,\mathbb{E}\{\exp[(v+\Lambda'\tilde{b}(u))'z_{t+1}] \mid \underline{w_t}\,,\underline{z_t}] \\
&= \exp\left[a(u)\,w_t + a_z(v+\Lambda'\tilde{b}(u),\pi)'z_t\right]\ .
\end{aligned}
$$

Example I.15 (Switching regimes Gaussian $AR(1)$ / extends Example I.7)

- *Notation:* $a(u) = u\,\rho$, $\tilde{b}(u)' = \left(u, \dfrac{u^2}{2}\right)$ *and* $\lambda' = (\nu, \sigma^2)$. *Thus,* $m = 2$.
- $\Lambda = \begin{bmatrix} \nu' \\ \sigma^{2'} \end{bmatrix}$ *is a* $(2, J)$*-matrix, where* $\nu = (\nu_1, \ldots, \nu_J)'$ *and* $\sigma^2 = (\sigma_1^2, \ldots, \sigma_J^2)'$.
- *The dynamics of* w_{t+1}, *conditionally to* $\underline{w_t}$, $\underline{z_{t+1}}$, *is given by:*

$$w_{t+1} = \nu' z_{t+1} + \rho\, w_t + (\sigma^{2'}\, z_{t+1})^{1/2} \varepsilon_{t+1}\,,$$

where ε_{t+1} *is a Gaussian white noise distributed as* $\mathcal{N}(0,1)$.
- *The joint conditional Laplace transform is:*

$$\begin{aligned} \varphi_t(u,v) &= \exp\left[a(u)\, w_t + a_z(v + \Lambda' \tilde{b}(u), \pi)' z_t\right] \\ &= \exp\left[u\,\rho\, w_t + a_z\left(v + u\,\nu + \frac{u^2}{2}\,\sigma^2, \pi\right)' z_t\right] \end{aligned}$$

I.4 – AFFINE (OR Car) PROCESSES

e. Switching regimes univariate Car(p) processes

- We follow the same strategy as in the previous section, but now, at the second step, we consider a univariate Index-Car(p) process with a conditional Laplace transform given by $\exp\left[a(u)\beta^{'} W_t + b(u)\right]$.
- We assume that $b(u)$ can be written:

$$\begin{aligned} b(u) &= \tilde{b}(u)'\lambda \quad \text{where} \\ \tilde{b}(u) &= (b_1(u), \ldots, b_m(u))' \text{ and } \lambda = (\lambda_1, \ldots, \lambda_m)'. \end{aligned} \tag{I.5}$$

- At the third step, we assume that the parameters λ_i are stochastic and linear functions of $Z_t = (z_t', \ldots, z_{t-p}')'$. More precisely, we assume that the conditional distribution of w_{t+1} given $\underline{w_t}$ and $\underline{z_{t+1}}$ has a Laplace transform given by:

$$\mathbb{E}[\exp(uw_{t+1})|\, \underline{w_t}\,, \underline{z_{t+1}}] = \exp\left[a(u)\beta^{'} W_t + \tilde{b}(u)^{'}\Lambda Z_t\right] ,$$

where Λ is a $[m, (p+1)J]$ matrix.

I.4 – AFFINE (OR Car) PROCESSES

e. Switching regimes univariate Car(p) processes

- We assume no instantaneous causality between w_{t+1} and z_{t+1} and we admit one more lag in Z_t than in W_t.

- This assumption is useful at the pricing level in order to obtain simple (explicit) pricing formulas with an exponential-affine stochastic discount factor [see Dai, Singleton and Yang (2007), Monfort and Pegoraro (2007)].

- The conditional Laplace transform of $(w_{t+1}, z'_{t+1})'$ given $\underline{w_t}$, $\underline{z_t}$ has the following form:

$$\begin{aligned} & \mathbb{E}\left[\exp(uw_{t+1} + v'z_{t+1}) \,|\, \underline{z_t},\, \underline{w_t}\right] \\ = \; & \exp\left\{a(u)\beta' W_t + \left[e'_1 \otimes a_z(v,\pi)' + \tilde{b}(u)'\Lambda\right] Z_t\right\}\,, \end{aligned}$$

where e_1 is the first component of the canonical basis in $\mathbb{R}^{p+1}$, and where $\otimes$ denotes the Kronecker product. The process $(w_{t+1}, z'_{t+1})'$ is thus Car($p+1$).

Example I.16 (Switching regimes Gaussian $AR(p)$ / extends Example I.13)

- *Let us start from the AR(p) model. Its conditional Laplace transform is given by:*

$$E\left[\exp(uw_{t+1})\,|\,\underline{w_t}\right] = \exp\left[u\varphi' W_t + u\nu + \frac{\sigma^2}{2}u^2\right],$$

and the function $b(u)$ has the form (I.5) with $\tilde{b}(u)' = \left(u, \frac{u^2}{2}\right)$ and $\lambda' = (\nu, \sigma^2)$.

- *If λ is replaced by ΛZ_t, the joint process $(w_{t+1}, z'_{t+1})'$ is Car($p+1$) with a conditional Laplace transform given by:*

$$\begin{aligned} &\mathbb{E}\left[\exp(uw_{t+1} + v'z_{t+1})\,|\,\underline{z_t},\,\underline{w_t}\right] \\ = \quad &\exp\left[u\varphi' W_t + \left(u, \frac{u^2}{2}\right)\Lambda Z_t + a_z(v,\pi)z_t\right]. \end{aligned}$$

Example I.16 (Switching regimes Gaussian $AR(p)$ / extends Example I.13)

- *More precisely, the dynamics is given by [using the notation* $\Lambda = \begin{pmatrix} \lambda_1' \\ \lambda_2' \end{pmatrix}$*]:*

$$w_{t+1} = \lambda_1' Z_t + \varphi' W_t + (\lambda_2' Z_t)^{1/2} \varepsilon_{t+1} \,,$$

where ε_{t+1} *is a Gaussian white noise distributed as* $\mathcal{N}(0, \sigma^2)$*,* $Z_t = (z_t', \ldots, z_{t-p}')'$ *and* z_t *is a homogeneous Markov chain.*

- *In particular, let us consider the case:*

$$\Lambda = \begin{bmatrix} (1, -\varphi_1, \ldots, -\varphi_p) \otimes \nu^{*'} \\ e_1' \otimes \sigma^{*2'} \end{bmatrix}$$

and $\nu^{*'} = (\nu_1^*, \ldots, \nu_J^*)$*,* $\sigma^{*2'} = (\sigma_1^{*2}, \ldots, \sigma_J^{*2})$*; the cond. distribution of* w_{t+1} *given* $\underline{w_t}$ *and* $\underline{z_{t+1}}$ *is the one corresponding to the switching AR(p) model defined by:*

$$\begin{aligned} w_{t+1} - \nu^{*'} z_t \quad &= \quad \varphi_1 \left(w_t - \nu^{*'} z_{t-1} \right) + \ldots + \varphi_p \left(w_{t+1-p} - \nu^{*'} z_{t-p} \right) \\ &\quad + (\sigma^{*'} z_t) \varepsilon_{t+1} \,. \end{aligned}$$

Example I.17 (Switching regimes $ARG(p)$ / extends Example I.14)

- *Let us now start from the ARG(p) process associated with the conditional Laplace transform :*

$$\mathbb{E}\left[\exp(uw_{t+1}) \,|\, \underline{w_t}\right] = \exp\left[\tfrac{u}{1-u\mu}\varphi' W_t - \nu \log(1-u\mu)\right] .$$

- *Here we have* $\tilde{b}(u) = -\log(1-u\mu)$ *and* $\lambda = \nu$. *If* ν *is replaced by* ΛZ_t, *where* $\Lambda Z_t > 0$, *the process* w_t *has, conditionally to the process* z_t, *a semi-strong AR(p) representation given by:*

$$w_{t+1} = \mu\Lambda Z_t + \varphi_1 w_t + \ldots + \varphi_p w_{t+1-p} + \zeta_{t+1} ,$$

where ζ_{t+1} *is a conditionally heteroscedastic martingale difference. For instance, we can take :*

$$\Lambda = e_1' \otimes \frac{\tilde{\nu}\,'}{\mu} ,$$

where $\tilde{\nu}\,' = (\tilde{\nu}_1, \ldots, \tilde{\nu}_J)$, $\tilde{\nu}_j \geq 0$. *We have* $\Lambda Z_t = \frac{\tilde{\nu}}{\mu}' z_t$ *and, condit. to the process* z_t, *the process* w_t *has a semi-strong AR(p) representation given by:*

$$w_{t+1} = \tilde{\nu}\,' z_t + \varphi_1 w_t + \ldots + \varphi_p w_{t+1-p} + \zeta_{t+1} .$$

Example I.17 (Switching regimes $ARG(p)$ / extends Example I.14)

- *It is also possible to consider a Λ of the form:*

$$(1, -\varphi_1, \dots, -\varphi_p) \otimes \frac{\tilde{\nu}'}{\mu}$$

if $\min(\tilde{\nu}_i) > \max(\tilde{\nu}_i) \sum_{i=1}^{J} \varphi_j$.

- *Indeed, in this case we have:*

$$\Lambda Z_t = \frac{1}{\mu}\left(\tilde{\nu}' z_t - \sum_{i=1}^{J} \varphi_j \tilde{\nu}' z_{t-i}\right) \geq 0$$

.

- *The weak conditional AR(p) representation is then given by:*

$$\begin{aligned} w_{t+1} - \tilde{\nu}' z_t &= \varphi_1 (w_t - \tilde{\nu}' z_{t-1}) + \dots + \varphi_p (w_{t+1-p} - \tilde{\nu}' z_{t-p}) \\ &\quad + \zeta_{t+1} . \end{aligned}$$

f. Stochastic parameters univariate affine processes

- Let us consider a univariate Car(p) process characterized by the following conditional Laplace transform:

$$\mathbb{E}_t\left[\exp(uw_{t+1})\right] = \exp[a_0(u)'W_t + \widetilde{b}_0(u)'\delta]$$

  where $W_t = [w_t, \ldots, w_{t+1-p}]'$, $\delta = [\delta_1, \ldots, \delta_m]'$.

- This dynamics can be generalized to the case of *stochastic parameters* by replacing δ with $\Delta\,\ell_{t+1}$, where ℓ_{t+1} is a k-dim. stochastic process and where Δ is a (m, k)-matrix.
- In this case, if ℓ_{t+1} is a Car(1) process with conditional Laplace transform:

$$\mathbb{E}[\exp(v'\ell_{t+1}) \,|\, \underline{w}_t, \underline{\ell}_t] = \exp[a_1(v)'\ell_t + b_1(v)]$$

f. Stochastic parameters univariate affine processes

- Then, $y_{t+1} = (w_{t+1}, \ell'_{t+1})'$ is a Car(p) process:

$$\mathbb{E}[\exp(uw_{t+1} + v'\ell_{t+1}) \mid \underline{w}_t, \underline{\ell}_t]$$

$$= \mathbb{E}\{\exp(v'\ell_{t+1})\mathbb{E}[\exp(uw_{t+1}) \mid \underline{w}_t, \underline{\ell}_{t+1}] \mid \underline{w}_t, \underline{\ell}_t\}$$

$$= \mathbb{E}\{\exp[a_0(u)'W_t + \widetilde{b}_0(u)'\Delta\ell_{t+1} + v'\ell_{t+1}] \mid \underline{w}_t, \underline{\ell}_t\}$$

$$= \exp\{a_0(u)'W_t + a_1[\Delta'\widetilde{b}_0(u) + v]'\ell_t + b_1[\Delta'\widetilde{b}_0(u) + v]\}$$

Example I.18 (Gaussian $AR(p)$ / extends Example I.13)

- *Notations:* $b_0(u) = \left(u, \ \frac{u^2}{2}\right)'$ *and* $\delta = (\nu, \sigma^2)' \in \mathcal{D} = \mathbb{R} \times \mathbb{R}^+$.
- *The vector* $\delta = (\delta_1, \delta_2)' = (\nu, \sigma^2)'$ *can be replaced by* $\Delta\, \ell_{t+1}$*, where* $\ell_{1,t+1}$ *and* $\ell_{2,t+1}$ *are independent* $AR(1)$ *and* $ARG(1)$ *processes (stochastic mean and stochastic volatility).*

Example I.19 (ARG(p) model / extends Example I.14)

- $b_0(u) = -\nu \log(1 - u\mu)$, $\delta = \nu$.
- $\nu > 0$ *can be specified as a scalar ARG process* ℓ_{t+1}.

I.4 – AFFINE (OR Car) PROCESSES

g. Building Multivariate Affine Processes

- Let us present the recursive approach to build multivariate Car(1) and associated Index-Car(p) processes with conditionally correlated scalar components.
- For ease of exposition, we will consider the bivariate case. All results can be easily extended to the general n-dimensional case.
- First, we consider two univariate exponential-affine Laplace transforms:

$$\begin{aligned} & \exp\left[a_1(u_1)x_{1,t} + b_1(u_1)\right], \\ \text{and} \qquad & \exp\left[a_2(u_2)x_{2,t} + b_2(u_2)\right], \end{aligned} \tag{I.6}$$

  characterizing the conditional distribution of two univariate stochastic processes $(x_{1,t})$ and $(x_{2,t})$.

I.4 – AFFINE (OR Car) PROCESSES

g. Building Multivariate Affine Processes

- Second, let us consider the bivariate process (w_t), with $w_t = (w_{1,t}, w_{2,t})'$, and let us assume that the conditional distribution of $w_{1,t+1}$ given $(w_{2,t+1}, \underline{w_{1,t}}, \underline{w_{2,t}})$ has a Laplace transform given by:

$$\begin{aligned} &\mathbb{E}_t[\exp(u_1 w_{1,t+1}) \mid w_{2,t+1}, \underline{w_{1t}}, \underline{w_{2t}}] \\ &= \exp\left[a_1(u_1)(\beta_o w_{2,t+1} + \beta_{11} w_{1,t} + \beta_{12} w_{2,t}) + b_1(u_1)\right] \end{aligned} \tag{I.7}$$

- and that the conditional distribution of $w_{2,t+1}$, given $(\underline{w_{1,t}}, \underline{w_{2,t}})$, has the following Laplace transform:

$$\begin{aligned} &\mathbb{E}_t[\exp(u_2 w_{2,t+1}) \mid \underline{w_{1,t}}, \underline{w_{2,t}}] \\ &= \exp\left[a_2(u_2)(\beta_{21} w_{1,t} + \beta_{22} w_{2,t}) + b_2(u_2)\right] . \end{aligned} \tag{I.8}$$

- Note that, if the Laplace transforms (I.6) correspond to positive stochastic processes and if the parameters $\beta_o, \beta_{11}, \beta_{12}, \beta_{21}, \beta_{22}$ are positive, then the bivariate process (w_t) has positive components.

g. Building Multivariate Affine Processes

- (w_t) is a bivariate Car(1) process with conditional Laplace transform :

$$\begin{aligned}
& \mathbb{E}[\exp(u_1 w_{1,t+1} + u_2 w_{2,t+1}) | \underline{w_{1,t}}, \underline{w_{2,t}}] \\
= \quad & \exp\Big\{[a_1(u_1)\beta_{11} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{21}]w_{1,t} \\
& +[a_1(u_1)\beta_{12} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{22}]w_{2,t} + b_1(u_1) + b_2(u_2 + a_1(u_1)\beta_o)\Big\} .
\end{aligned} \tag{I.9}$$

- **Proof:**

$$\begin{aligned}
& \mathbb{E}[\exp(u_1 w_{1,t+1} + u_2 w_{2,t+1}) | \underline{w_{1,t}}, \underline{w_{2,t}}] \\
= \quad & \mathbb{E}\left[\exp(u_2 w_{2,t+1}) \mathbb{E}\left(\exp(u_1 w_{1,t+1}) | \underline{w_{1,t}}, \underline{w_{2,t+1}}\right) \ | \underline{w_{1,t}}, \underline{w_{2,t}}\right] \\
= \quad & \exp\left[a_1(u_1)(\beta_{11} w_{1,t} + \beta_{12} w_{2,t}) + b_1(u_1)\right] \mathbb{E}_t\left[(u_2 + a_1(u_1)\beta_o) w_{2,t+1} \mid \underline{w_{1,t}}, \underline{w_{2,t}}\right] \\
= \quad & \exp\Big[a_1(u_1)(\beta_{11} w_{1,t} + \beta_{12} w_{2,t}) + b_1(u_1) \\
& +a_2(u_2 + a_1(u_1)\beta_o)(\beta_{21} w_{1,t} + \beta_{22} w_{2,t}) + b_2(u_2 + a_1(u_1)\beta_o)\Big] \\
= \quad & \exp\Big\{[a_1(u_1)\beta_{11} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{21}]w_{1,t} \\
& +[a_1(u_1)\beta_{12} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{22}]w_{2,t} + b_1(u_1) + b_2(u_2 + a_1(u_1)\beta_o)\Big\} .
\end{aligned}$$

Example I.20 ($VARG(1)$ process / extends Example I.11)

- *The bivariate $VARG(1)$ process (w_t) (say), with $w_t = (w_{1,t}, w_{2,t})'$, is specified assuming that $(x_{1,t})$ and $(x_{2,t})$ are univariate $ARG(1)$ processes with conditional Laplace transforms (I.6) given by:*

$$\begin{aligned} a_1(u_1) &= \frac{\rho_1 u_1}{1 - u_1\mu_1}\,,\; b_1(u_1) = -\nu_1 \log(1 - u_1\mu_1) \\ a_2(u_2) &= \frac{\rho_2 u_2}{1 - u_2\mu_2}\,,\; b_2(u_2) = -\nu_2 \log(1 - u_2\mu_2)\,, \end{aligned} \tag{I.10}$$

  and, therefore, its conditional Laplace transform is given by (I.9), with functions $a_1(u_1)$, $b_1(u_1)$, $a_2(u_2)$ and $b_2(u_2)$ defined in I.11.

- *The conditional p.d.f of $w_{2,t+1}$, given $(\underline{w_{1,t}}, \underline{w_{2,t}})$, is given by*

$$f(w_{2,t+1} \,|\, \beta_{21} w_{1,t} + \beta_{22} w_{2,t}; \mu_2, \nu_2, \rho_2),$$

- *The conditional p.d.f. of $w_{1,t+1}$, given $(\underline{w_{1,t}}, \underline{w_{2,t+1}})$, is*

$$f(w_{1,t+1} \,|\, \beta_o w_{2,t+1} + \beta_{11} w_{1,t} + \beta_{12} w_{2,t}; \mu_1, \nu_1, \rho_1)\,.$$

Example I.20 ($VARG(1)$ process / extends Example I.11)

- *Therefore, the conditional p.d.f. of w_{t+1}, given w_t, is defined as:*

$$
\begin{aligned}
& f\left(w_{t+1} \mid w_t; \mu, \nu, \rho\right) \\
& \qquad = f(w_{1,t+1} \mid \beta_o w_{2,t+1} + \beta_{11} w_{1,t} + \beta_{12} w_{2,t}; \mu_1, \nu_1, \rho_1) \\
& \qquad\quad \times f(w_{2,t+1} \mid \beta_{21} w_{1,t} + \beta_{22} w_{2,t}; \mu_2, \nu_2, \rho_2)\,, \\
& \textit{notation: } \mu = (\mu_1, \mu_2)'\,,\ \nu = (\nu_1, \nu_2)'\,,\ \rho = (\rho_1, \rho_2)'\,.
\end{aligned}
$$

- *Now, if we introduce the notations $\varphi_o = \rho_1\beta_o$, $\varphi_{11} = \rho_1\beta_{11}$, $\varphi_{12} = \rho_1\beta_{12}$, $\varphi_{21} = \rho_2\beta_{21}$, $\varphi_{22} = \rho_2\beta_{22}$, we can write the conditional means and conditional variances of $(w_{1,t+1}, w_{2,t+1})'$, given w_t, as follows:*

$$
\begin{aligned}
\mathbb{E}(w_{1,t+1} \mid w_t) &= \nu_1\mu_1 + \varphi_o\nu_2\mu_2 + (\varphi_{11} + \varphi_o\varphi_{21})w_{1,t} + (\varphi_{12} + \varphi_o\varphi_{22})w_{2,t} \\
\mathbb{E}(w_{2,t+1} \mid w_t) &= \nu_2\mu_2 + \varphi_{21} w_{1,t} + \varphi_{22} w_{2,t} \\
\mathbb{V}ar(w_{1,t+1} \mid w_t) &= \nu_1\mu_1^2 + 2\varphi_o\nu_2\mu_1\mu_2 \\
&\quad +2\mu_1[(\varphi_{11} + \varphi_o\varphi_{21})w_{1,t} + (\varphi_{12} + \varphi_o\varphi_{22})w_{2,t}] \\
\mathbb{V}ar(w_{2,t+1} \mid w_t) &= \nu_2\mu_2^2 + 2\mu_2[\varphi_{21} w_{1,t} + \varphi_{22} w_{2,t}]\,,
\end{aligned}
$$

Example I.20 ($VARG(1)$ process / extends Example I.11)

- *and therefore, the process (w_t) has the following semi-strong VAR(1) representation:*

$$\begin{bmatrix} w_{1,t+1} \\ w_{2,t+1} \end{bmatrix} = \begin{bmatrix} \nu_1\mu_1 + \varphi_o\nu_2\mu_2 \\ \nu_2\mu_2 \end{bmatrix} + \begin{bmatrix} (\varphi_{11} + \varphi_o\varphi_{21}) & (\varphi_{12} + \varphi_o\varphi_{22}) \\ \varphi_{21} & \varphi_{22} \end{bmatrix} \begin{bmatrix} w_{1,t} \\ w_{2,t} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,t+1} \\ \varepsilon_{2,t+1} \end{bmatrix} \tag{I.11}$$

- *where (ε_t), with $\varepsilon_t = (\varepsilon_{1,t}, \varepsilon_{2,t})'$, is a bivariate conditionally heteroskedastic martingale difference, with:*

$$\mathbb{V}ar(\varepsilon_{1,t+1} \mid \varepsilon_t) = \mathbb{V}ar(w_{1,t+1} \mid w_t)$$

$$\mathbb{V}ar(\varepsilon_{2,t+1} \mid \varepsilon_t) = \mathbb{V}ar(w_{2,t+1} \mid w_t)$$

$$\mathbb{C}ov(\varepsilon_{1,t+1}, \varepsilon_{2,t+1} \mid \varepsilon_t) = \mathbb{C}ov(w_{1,t+1}, w_{2,t+1} \mid w_t) = 0$$

- *Denoting Φ the autoregressive matrix in (I.11), the VARG(1) process (w_t) is stationary if and only if $\det(I_{2\times 2} - \Phi z) \neq 0$ for all $|z| \leq 1$ and, in this case, its marginal mean is $\mathbb{E}(w_t) = (I_{2\times 2} - \Phi)^{-1}\bar{\nu}$, with $\bar{\nu} = (\nu_1\mu_1 + \varphi_o\nu_2\mu_2, \nu_2\mu_2)'$.*

Example I.20 ($VARG(1)$ process / extends Example I.11)

- *The marginal variances of $(\varepsilon_{1,t})$ and $(\varepsilon_{2,t})$ are respectively given by:*

$$\begin{aligned}\mathbb{V}ar(\varepsilon_{1,t}) &= \nu_1\mu_1^2 + 2\varphi_o\nu_2\mu_1\mu_2 + 2\mu_1(\varphi_{11} + \varphi_o\varphi_{21}, \varphi_{12} + \varphi_o\varphi_{22})'(I_{2\times 2} - \Phi)^{-1}\bar{\nu} \\ \mathbb{V}ar(\varepsilon_{2,t}) &= \nu_2\mu_2^2 + 2\mu_2(\varphi_{21}, \varphi_{22})'(I_{2\times 2} - \Phi)^{-1}\bar{\nu}\,,\end{aligned} \quad (I.12)$$

  and, clearly, $\mathbb{C}ov(\varepsilon_{1,t}, \varepsilon_{2,t}) = 0$.

- *Moreover, the marginal variance of (w_t) is such that:*

$$vec[\mathbb{V}ar(w_t)] = (I_{4\times 4} - \Phi\otimes\Phi)^{-1}vec[\mathbb{V}ar(\varepsilon_t)].$$

I.4 – AFFINE (OR Car) PROCESSES

g. Building Multivariate Affine Processes

- We consider now a bivariate process (w_t), with $w_t = (w_{1,t}, w_{2,t})'$, and we introduce the following notations:

$$W_{1t} = (w_{1,t}, \dots, w_{1,t+1-p})' \, , W_{2t} = (w_{2,t}, \dots, w_{2,t+1-p})'$$

.

- A bivariate Index-Car(p) process (w_t), associated to a bivariate Car(1) process with conditional Laplace transform (I.9), is defined by the conditional Laplace transforms:

$$\mathbb{E}_t[\exp(u_1 w_{1,t+1}) \,|\, w_{2,t+1}, \underline{w_{1,t}}, \underline{w_{2,t}}] = \exp\left[a_1(u_1)(\beta_o w_{2,t+1} + \beta_{11}' \, W_{1t} + \beta_{12}' \, W_{2t}) + b_1(u_1)\right] \, ,$$

$$\mathbb{E}_t[\exp(u_2 w_{2,t+1}) \,|\, \underline{w_{1,t}}, \underline{w_{2,t}}] = \exp\left[a_2(u_2)(\beta_{21}' \, W_{1t} + \beta_{22}' \, W_{2t}) + b_2(u_2)\right] \, , \tag{I.13}$$

where β_{ij}, for each $i, j \in \{1, 2\}$, are p-vectors.

I.4 – AFFINE (OR Car) PROCESSES

g. Building Multivariate Affine Processes

- The process (w_t) is a Car(p) process with a conditional Laplace transform given by:

$$\mathbb{E}\left[\exp(u' w_{t+1}) \mid \underline{w_t}\right] = \exp\{[a_1(u_1)\beta_{11} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{21}]' \, W_{1t} \\ +[a_1(u_1)\beta_{12} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{22}]' \, W_{2t} \\ + b_1(u_1) + b_2(u_2 + a_1(u_1)\beta_o)\} \, . \tag{I.14}$$

- From the properties of Car(p) processes [see Darolles, Gourieroux and Jasiak (2006) for details] we get a representation of the form:

$$\begin{cases} w_{1,t+1} = \alpha_1 + \alpha_o w_{2,t+1} + \alpha_{11}' \, W_{1t} + \alpha_{12}' \, W_{2t} + \varepsilon_{1,t+1} \\ w_{2,t+1} = \alpha_2 + \alpha_{21}' \, W_{1t} + \alpha_{22}' \, W_{2t} + \varepsilon_{2,t+1} \end{cases} \tag{I.15}$$

where the error terms satisfy :

$$\begin{aligned} \mathbb{E}[\varepsilon_{1,t+1} \mid w_{2,t+1}, \underline{w_t}] &= 0 \\ \mathbb{E}[\varepsilon_{2,t+1} \mid \underline{w_t}] &= 0 \, . \end{aligned}$$

I.4 – AFFINE (OR Car) PROCESSES

g. Building Multivariate Affine Processes

- In particular, we get

$$
\begin{aligned}
\mathbb{E}[\varepsilon_{1,t+1} \mid \underline{w_t}] &= 0 \\
\mathbb{E}[\varepsilon_{2,t+1} \mid \underline{w_t}] &= 0 \\
\mathbb{C}ov(\varepsilon_{1,t+1}, \varepsilon_{2,t+1}) &= \mathbb{E}(\varepsilon_{1,t+1}\varepsilon_{2,t+1} \mid \underline{w_t}) \\
&= \mathbb{E}\left[\varepsilon_{2,t+1}\mathbb{E}(\varepsilon_{1,t+1} \mid w_{2,t+1}, \underline{w_t}) \mid \underline{w_t}\right] \\
&= 0\,.
\end{aligned} \tag{I.16}
$$

- So, the error terms are non correlated, cond. heteroskedastic, martingale differences.
- In particular, in the stationary case, $(\varepsilon_{1,t})$ and $(\varepsilon_{2,t})$ are uncorr. (semi-strong) white noises and (I.15) is a semi-strong recursive VAR(p) representation of the process (w_t).

Example I.21 (Gaussian $VAR(p)$ process / extends Example I.8)

- *In this case the conditional distributions defined by (I.6) are Gaussian, with affine expectations and fixed variances. In other words:*

$$\begin{aligned} a_1(u_1) &= \rho_1 u_1 \,, \; b_1(u_1) = \nu_1 u_1 + \frac{\sigma_1^2 u_1^2}{2} \\ a_2(u_2) &= \rho_2 u_2 \,, \; b_2(u_2) = \nu_2 u_2 + \frac{\sigma_2^2 u_2^2}{2} \,. \end{aligned} \tag{I.17}$$

- *Using the notations $\varphi_o = \rho_1 \beta_o$, $\varphi_{11} = \rho_1 \beta_{11}$, $\varphi_{12} = \rho_1 \beta_{12}$, $\varphi_{21} = \rho_2 \beta_{21}$, $\varphi_{22} = \rho_2 \beta_{22}$, we have the following strong VAR(p) recursive representation for the process $w_t = (w_{1,t}, w_{2,t})'$:*

$$\left\{ \begin{array}{lcl} w_{1,t+1} & = & \nu_1 + \varphi_o w_{2,t+1} + \varphi_{11}' W_{1t} + \varphi_{12}' W_{2t} + \sigma_1 \eta_{1,t+1} \\ w_{2,t+1} & = & \nu_2 + \varphi_{21}' W_{1t} + \varphi_{22}' W_{2t} + \sigma_2 \eta_{2,t+1} \,, \end{array} \right. \tag{I.18}$$

where $\eta_t = (\eta_{1,t}, \eta_{2,t})'$ is a bivariate Gaussian white noise distributed as $\mathcal{N}(0, I_2)$, where I_2 denotes the (2×2) identity matrix.

Example I.22 ($VARG(p)$ process / extends Example I.20)

- *Once the VARG(1) process is specified, the associated Index-Car(p) process (w_t) (say), with $w_t = (w_{1,t}, w_{2,t})'$ and named Vector ARG(p) process, is characterized by the conditional Laplace transform (I.14).*
- *Moreover, by using the notations $\varphi_o = \rho_1 \beta_o$, $\varphi_{11} = \rho_1 \beta_{11}$, $\varphi_{12} = \rho_1 \beta_{12}$, $\varphi_{21} = \rho_2 \beta_{21}$, $\varphi_{22} = \rho_2 \beta_{22}$, the process (w_t) has a semi-strong VAR(p) representation:*

$$\begin{cases} w_{1,t+1} = \nu_1 \mu_1 + \varphi_o w_{2,t+1} + \varphi_{11}' \, W_{1t} + \varphi_{12}' \, W_{2t} + \xi_{1,t+1} \\ w_{2,t+1} = \nu_2 \mu_2 + \varphi_{21}' \, W_{1t} + \varphi_{22}' \, W_{2t} + \xi_{2,t+1} \, , \end{cases} \tag{I.19}$$

where $\xi_{1,t}$ and $\xi_{2,t}$ are non correlated, conditionally heteroskedastic, martingale differences.

- *The conditional variances of $\xi_{1,t+1}$ and $\xi_{2,t+1}$ are given by:*

$$\begin{aligned} \mathbb{V}[\xi_{1,t+1} \mid \underline{w_t}] &= \nu_1 \mu_1^2 + 2\mu_1 [\varphi_o (\nu_2 \mu_2 + \varphi_{21}' \, W_{1t} + \varphi_{22}' \, W_{2t}) + \varphi_{11}' \, W_{1t} + \varphi_{12}' \, W_{2t}] \\ \mathbb{V}[\xi_{2,t+1} \mid \underline{w_t}] &= \nu_2 \mu_2^2 + 2\mu_2 (\varphi_{21}' \, W_{1t} + \varphi_{22}' \, W_{2t}) \, . \end{aligned} \tag{I.20}$$

- *We are able to build multivariate positive Car processes with instant. causality among scalar components and a (exact) likelihood function known in closed form. In continuous time, with multivariate CIR processes, the latter property is obtained only in the case of indep. among factors.*

I.4 – AFFINE (OR Car) PROCESSES

h. Wishart autoregressive (WAR) processes

- WAR are valued in the space of $(L \times L)$ symmetric positive definite matrices. Let Y_{t+1} be a $WAR_L(K, M, \Omega)$ process. It is defined by:

$$\begin{aligned} & \mathbb{E}[\exp\ Tr(\Gamma Y_{t+1}) | \underline{Y_t}] \\ = \quad & \exp\left\{ Tr[M'\Gamma(Id - 2\Omega\Gamma)^{-1} M Y_t] - \frac{K}{2} \log[det(Id - 2\Omega\Gamma)] \right\}, \end{aligned} \tag{I.21}$$

where:

  - Γ is (without loss of generality) a symmetric matrix; $Tr(\Gamma Y_{t+1})$ is equal to:

$$\sum_{i=1}^{L} (\Gamma Y_{t+1})_{ii} = \sum_{i=1}^{L} \sum_{j=1}^{L} \gamma_{ij} Y_{t+1,ij} = \sum_{i=1}^{L} \gamma_{ii} Y_{t+1,ii} + \sum_{i<j}^{L} (\gamma_{ij} + \gamma_{ji}) Y_{t+1,ij}.$$

  - K is a positive scalar (degree of freedom parameter),
  - M is a $(L \times L)$ matrix (of AR parameters),
  - Ω is a $(L \times L)$ symmetric positive definite matrix.
  - Γ has to be such that $\|\Omega^{1/2}\,\Gamma\,\Omega^{1/2}\| < 1$ to ensure that the Laplace transform is well defined (The norm of a symmetric matrix is equal to its largest eigenvalue).

- If K integer, Y_{t+1} obtained from:

$$\begin{cases} Y_{t+1} & = & \sum_{k=1}^{K} x_{k,t+1} x'_{k,t+1} \\ x_{k,t+1} & = & M x_{k,t} + \varepsilon_{k,t+1}, \quad k \in \{1, \ldots, K\}, \end{cases}$$

where $\varepsilon_{k,t+1} \sim i.i.d. \mathcal{N}(0, \Omega)$ (independent across k's).

- We have:

$$Y_{t+1} = M\, Y_t\, M' + K\Omega, + \eta_{t+1}$$

where η_{t+1} is a matrix martingale difference, i.e. Y_t follows a matrix semi-strong AR(1)-like process.

Proof. See next slide.

- Particular case: $L = 1$ (univariate).

$$\mathbb{E}[\exp(uY_{t+1}) \,|\, \underline{Y_t}] = \exp\left[\frac{um^2}{1 - 2\omega u} Y_t - \frac{K}{2} \log(1 - 2\omega u)\right]$$

$\Leftrightarrow$ $ARG(1)$ process (Example I.11) with $\rho = m^2$, $\mu = 2\omega$, $\nu = \frac{K}{2}$.

Proof: L.T. of a $WAR_L(K, M, \Omega))$ process (K integer).

- Let us consider $K = 1$. In this case we have $W_{t+1} = x_{t+1}\, x'_{t+1}$ with $x_{t+1} = Mx_t + \Omega^{1/2} u_{t+1}$ and $u_{t+1} \sim i.i.d.\mathcal{N}(0, Id_L)$.

- We have:

$$\mathbb{E}[\exp(Tr\Gamma W_{t+1})|\underline{w_t}] = \mathbb{E}\{\mathbb{E}[\exp(Tr\Gamma x_{t+1}x'_{t+1})|\underline{x}_t]|\underline{w_t}\}$$

and:

$$\begin{aligned}
& \mathbb{E}[\exp(Tr\Gamma x_{t+1}x'_{t+1})|\underline{x}_t] = \mathbb{E}[\exp(x'_{t+1}\Gamma x_{t+1}|\underline{x}_t] \\
= \quad & \mathbb{E}[\exp(Mx_t + \Omega^{1/2}u_{t+1})'\Gamma(Mx_t + \Omega^{1/2}u_{t+1})/x_t] \\
= \quad & \exp(x'_t M'\Gamma M x_t)\mathbb{E}[\exp(2x'_t M'\Gamma\Omega^{1/2}u_{t+1} + u'_{t+1}\Omega^{1/2}\Gamma\Omega^{1/2}u_{t+1})/x_t] \\
= \quad & \frac{exp(x'_t M'\Gamma M x_t)}{(2\pi)^{L/2}} \times \\
& \int_{\mathbb{R}^L} \exp\left[2x'_t M'\Gamma\Omega^{1/2}u_{t+1} - u'_{t+1}\left(\frac{1}{2}Id_L - \Omega^{1/2}\Gamma\Omega^{1/2}\right)u_{t+1}\right] du_{t+1}.
\end{aligned}$$

Proof: L.T. of a $WAR_L(K, M, \Omega))$ process (K integer).

- Using Lemma 3 with

$$\mu' = 2x_t' M' \Gamma \Omega^{1/2}, Q = \frac{1}{2} Id_L - \Omega^{1/2} \Gamma \Omega^{1/2},$$

- and after some algebra, the RHS becomes:

$$\frac{\exp[x_t' M' \Gamma (Id_L - 2\,\Omega\,\Gamma)^{-1}\, M\, x_t]}{det[Id_L - 2\Omega^{1/2}\Gamma\Omega^{1/2}]} = \frac{\exp\, Tr[M' \Gamma (Id_L - 2\Omega^{-1}]\, M W_t]}{det[Id_L - 2\,\Omega\,\Gamma]^{1/2}},$$

which depends on x_t through W_t, and gives the result for $K = 1$.

- the result for any K integer follows.

I.4 – AFFINE (OR Car) PROCESSES

i. Switching Regimes Multivariate Affine Processes

- Switching regimes can be introduced in a multivariate Index - Car(p) model using a method extending the one retained in the univariate case.
- Let us assume that the functions $b_1(u_1)$, $b_2(u_2)$ appearing in (I.13) can be written, respectively, as $\widetilde{b_1}(u_1)'\lambda_1$ and $\widetilde{b_2}(u_2)'\lambda_2$.
- Let us replace λ_1 and λ_2, respectively by $\Lambda_1 Z_t$ and $\Lambda_2 Z_t$. Then, the Laplace transform of $(w_{1,t+1}, w_{2,t+1}, z_{t+1})$, condit. to $(\underline{w_{1,t}}, \underline{w_{2,t}}, \underline{z_t})$, has the form given in (I.22) and the process is Car($p+1$).
- Indeed, we obtain the following conditional Laplace transform of $(w_{1,t+1}, w_{2,t+1}, z_{t+1})$ given $(\underline{w_{1,t}}, \underline{w_{2,t}}, \underline{z_t})$:

$$\begin{aligned} & \mathbb{E}[\exp(u_1 w_{1,t+1} + u_2 w_{2,t+1} + v' z_{t+1}) \mid \underline{w_{1,t}}, \underline{w_{2,t}}, \underline{z_t}] \\ = \quad & \exp\Big\{[a_1(u_1)\beta_{11} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{21}]' W_{1t} \\ & +[a_1(u_1)\beta_{12} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{22}]' W_{2t} \\ & +[e_1' \otimes a_z(v,\pi)' + \widetilde{b_1}(u_1)'\Lambda_1 + \widetilde{b_2}(u_2 + a_1(u_1)\beta_o)'\Lambda_2] Z_t \Big\} , \end{aligned} \tag{I.22}$$

Example I.23 (Regime Switching Gaussian $VAR(p)$ process / extends Example I.21)

- *Let us take:*

$$
\begin{aligned}
a_1(u_1) &= \rho_1 u_1,\ \ b_1(u_1) = \nu_1 u_1 + \frac{\sigma_1^2}{2} u_1^2,\ \ \tilde{b_1}(u_1)' = \left(u_1, \frac{u_1^2}{2}\right), \\
a_2(u_2) &= \rho_2 u_2,\ \ b_2(u_2) = \nu_2 u_2 + \frac{\sigma_2^2}{2} u_2^2,\ \ \tilde{b_2}(u_2)' = \left(u_2, \frac{u_2^2}{2}\right), \\
\Lambda_1 &= \begin{pmatrix} \lambda_{11}' \\ \lambda_{12}' \end{pmatrix},\ \ \Lambda_2 = \begin{pmatrix} \lambda_{21}' \\ \lambda_{22}' \end{pmatrix},
\end{aligned}
$$

and let us adopt the notations $\varphi_o = \rho_1 \beta_o$, $\varphi_{11} = \rho_1 \beta_{11}$, $\varphi_{12} = \rho_1 \beta_{12}$, $\varphi_{21} = \rho_2 \beta_{21}$, $\varphi_{22} = \rho_2 \beta_{22}$.

- *We obtain the following Regime Switching Gaussian VAR(p) model:*

$$
\left\{
\begin{aligned}
w_{1,t+1} &= \lambda_{11}' Z_t + \varphi_o w_{2,t+1} + \varphi_{11}' W_{1t} + \varphi_{12}' W_{2t} + (\lambda_{12}' Z_t)^{1/2} \eta_{1,t+1} \\
w_{2,t+1} &= \lambda_{21}' Z_t + \varphi_{21}' W_{1t} + \varphi_{22}' W_{2t} + (\lambda_{22}' Z_t)^{1/2} \eta_{2,t+1}\ ,
\end{aligned}
\right.
\tag{I.23}
$$

where $\eta_t = (\eta_{1,t}, \eta_{2,t})'$ *is a Gaussian white noise distributed as* $\mathcal{N}(0, I_2)$, $Z_t = (z_t', \dots, z_{t-p}')'$, *and where* z_t *is a homogeneous J-state Markov chain.*

Example I.24 (Regime Switching $VARG(p)$ process / extends Example I.22)

- *If we take:*

$$a_1(u_1) = \frac{\rho_1 u_1}{1-u_1\mu_1}, \quad b_1(u_1) = -\nu_1 \log(1-u_1\mu_1), \quad \tilde{b_1}(u_1) = \log(1-u_1\mu_1),$$

$$a_2(u_2) = \frac{\rho_2 u_2}{1-u_2\mu_2}, \quad b_2(u_2) = -\nu_2 \log(1-u_2\mu_2), \quad \tilde{b_2}(u_2) = \log(1-u_2\mu_2)\,,$$

we obtain the (positive) Regime Switching VARG(p) model

$$\begin{cases} w_{1,t+1} = \mu_1\Lambda_1' Z_t + \varphi_o w_{2,t+1} + \varphi_{11}' W_{1t} + \varphi_{12}' W_{2t} + \xi_{1,t+1} \\ w_{2,t+1} = \mu_2\Lambda_2' Z_t + \varphi_{21}' W_{1t} + \varphi_{22}' W_{2t} + \xi_{2,t+1}\,, \end{cases} \tag{I.24}$$

where $\xi_{1,t}$ and $\xi_{2,t}$ are non correlated, conditionally heteroscedastic, martingale differences, the conditional variances being:

$$\begin{aligned} \mathbb{V}ar[\xi_{1,t+1} \mid \underline{w_t}] &= \Lambda_1' Z_t \mu_1^2 \\ &\quad +2\mu_1[\varphi_o(\Lambda_2' Z_t \mu_2 + \varphi_{21}' W_{1t} + \varphi_{22}' W_{2t}) + \varphi_{11}' W_{1t} + \varphi_{12}' W_{2t}] \\ \mathbb{V}ar[\xi_{2,t+1} \mid \underline{w_t}] &= \Lambda_2' Z_t \mu_2^2 + 2\mu_2(\varphi_{21}' W_{1t} + \varphi_{22}' W_{2t})\,. \end{aligned} \tag{I.25}$$

I.4 – AFFINE (OR Car) PROCESSES

j. Multivariate Stochastic Parameters Processes

- Consider the same framework as in Section f (Slide 50) when y_t is a n-dimensional vector. The same proof shows that $y_{t+1} = (w'_{t+1}, \ell'_{t+1})'$ is affine.

Example I.25 (Stochastic parameters Gaussian $VAR(1)$ / extends Example I.8)

Using the same notations as in Example I.8 and in Section f (Slide 50), we have

$$b_0(u) = \left(u', \frac{1}{2}(u \otimes u)'\right)' \quad \text{and} \quad \delta = (\mu', vec(\Sigma)')' \in \mathbb{R}^n \times vec(\mathcal{S}),$$

where $\mathcal{S}$ is the set of symmetric positive semi-definite matrices; μ can be replaced by a Gaussian VAR(1) process (frequently, a RW) and Σ by a Wishart Autoregressive process. In this case we talk about a Time-Varying Parameter (TVP) VAR model with/without Stochastic Volatility (SV).

I.4 – AFFINE (OR Car) PROCESSES

k. First Key Property: Multi-Horizon Laplace Transform

- w_{t+1} is multivariate affine of order 1 (contains the order p case, see Remark I.1), with functions $a(.)$, $b(.)$:

$$\mathbb{E}\left[\exp(u' w_{t+1}) \,\middle|\, \underline{w_t}\right] = \exp[a(u)' w_t + b(u)]. \qquad \text{(I.26)}$$

- Here we consider $u \in \mathbb{R}^n$ and functions a and b that do not depend on time. However, tractable formulas can also be obtained with $u \in \mathbb{C}^n$ or when a and b deterministically depend on time.

Prop. I.1 (Multi-horizon Laplace transform)

If Equation (I.26) is satisfied, we have, for any horizon $h \geq 1$:

$$\begin{aligned} & \varphi_{t,h}(\gamma_1, \dots, \gamma_h) := \mathbb{E}[\exp(\gamma_1' w_{t+1} + \dots + \gamma_h' w_{t+h}) | \underline{w_t}] \\ = \; & \exp[A_h(\gamma_1, \dots, \gamma_h)' w_t + B_h(\gamma_1, \dots, \gamma_h)], \end{aligned} \qquad \text{(I.27)}$$

where functions A_h and B_h can be evaluated recursively using:

$$\left\{ \begin{array}{lcl} A_h(\gamma_1, \dots, \gamma_h) & = & a[\gamma_1 + A_{h-1}(\gamma_2, \dots, \gamma_h)], \\ B_h(\gamma_1, \dots, \gamma_h) & = & b[\gamma_1 + A_{h-1}(\gamma_2, \dots, \gamma_h)] + B_{h-1}(\gamma_2, \dots, \gamma_h), \end{array} \right. \qquad \text{(I.28)}$$

starting the recursions with $A_0 \equiv 0$ and $B_0 \equiv 0$.

k. First Key Property: Multi-Horizon Laplace Transform

Proof Equation (I.27) is satisfied for $h = 1$ by definition of functions a and b (Equation I.26). Suppose it is satisfied for $h - 1$, with $h \geq 1$. We have:

$$\begin{aligned} & \mathbb{E}_t[\exp(\gamma_1' w_{t+1} + \cdots + \gamma_h' w_{t+h})] \\ = & \mathbb{E}_t[\mathbb{E}_{t+1}[\exp(\gamma_1' w_{t+1} + \cdots + \gamma_h' w_{t+h})]] \\ = & \mathbb{E}_t[\exp(\gamma_1' w_{t+1} + A_{h-1}(\gamma_2, \ldots, \gamma_h)' w_{t+1} + B_{h-1}(\gamma_2, \ldots, \gamma_h))], \end{aligned}$$

which leads to the result.

- In many applications, we will have to use the previous proposition for several maturities. For instance, if we consider H maturities $h \in \{1, \ldots, H\}$, we face the following structure of vectors γ:

$$\begin{array}{lcllll} h = 1 & : & \gamma_1^{(1)} & & & \\ h = 2 & : & \gamma_1^{(2)} & \gamma_2^{(2)} & & \\ \vdots & & \vdots & & \ddots & \\ h = H & : & \gamma_1^{(H)} & \ldots & & \gamma_H^{(H)}. \end{array} \tag{I.29}$$

I.4 – AFFINE (OR Car) PROCESSES

k. First Key Property: Multi-Horizon Laplace Transform

- For each $i \in \{1, \dots, H\}$, the recursions given in (I.28) must be applied to the set $\{\gamma_1^{(i)}, \dots, \gamma_i^{(i)}\}$.
- This involves $H \times (H+1)/2$ evaluations of the functions a and b, which can become time-consuming if H is large, which happens when considering long horizons in a model with relatively high frequency.
- Fewer evaluations are necessary when the sequences of γ vectors exhibit the following *reverse-order* structure:

$$\begin{array}{lcllll} h=1 & : & u_1 & & & \\ h=2 & : & u_2 & u_1 & & \\ \vdots & & \vdots & \ddots & \ddots & \\ h=H & : & u_H & \dots & u_2 & u_1, \end{array} \tag{I.30}$$

which is typically the case in term structure applications, as will be illustrated below. The following proposition formalizes this reverse-order case; the proof follows directly from Proposition I.1.

I.4 – AFFINE (OR Car) PROCESSES

k. First Key Property: Multi-Horizon Laplace Transform

Prop. I.2 (Reverse-order multi-horizon Laplace transform)

If (I.26) is satisfied, then we have, for any sequence $\{u_1, \dots, u_H\}$ and any $h \in \{1, \dots, H\}$:

$$\begin{aligned} \varphi_{t,h}(u_h, \dots, u_1) &= \mathbb{E}_t[\exp(u_h' w_{t+1} + \dots + u_1' w_{t+h})] \\ &= \exp(A_h' w_t + B_h), \end{aligned} \tag{I.31}$$

where the vectors $A_1, \dots, A_H$ and the scalars $B_1, \dots, B_H$ are obtained with only one recursion:

$$\left\{ \begin{aligned} A_h &= a(u_h + A_{h-1}), \\ B_h &= b(u_h + A_{h-1}) + B_{h-1}, \end{aligned} \right. \tag{I.32}$$

starting from $A_0 \equiv 0$ and $B_0 \equiv 0$.

I.4 – AFFINE (OR Car) PROCESSES

k. First Key Property: Multi-Horizon Laplace Transform

Let us consider the case where functions a and b, which define the Laplace transform of w_t, deterministically depend on time:

$$\mathbb{E}_t\left[\exp\left(u' w_{t+1}\right)\right] = \exp\left[a_{t+1}(u)' w_t + b_{t+1}(u)\right]. \qquad (I.33)$$

Lemma 5

If the process w_t satisfies equation (I.33), then we have:

$$\begin{aligned} \varphi_{t,h}(\gamma_1,\ldots,\gamma_h) &:= \mathbb{E}_t\left[\exp\left(\gamma_1' w_{t+1} + \cdots + \gamma_h' w_{t+h}\right)\right] \\ &= \exp\left[A_{t,h}(\gamma_1,\ldots,\gamma_h)' w_t + B_{t,h}(\gamma_1,\ldots,\gamma_h)\right], \end{aligned} \qquad (I.34)$$

where the functions $A_{t,h}$ and $B_{t,h}$ can be computed using the following recursive relations:

$$\left\{\begin{array}{lcl} A_{t,h} &=& a_{t+1}\left[\gamma_1 + A_{t+1,h-1}(\gamma_2,\ldots,\gamma_h)\right], \\ B_{t,h} &=& b_{t+1}\left[\gamma_1 + A_{t+1,h-1}(\gamma_2,\ldots,\gamma_h)\right] + B_{t+1,h-1}(\gamma_2,\ldots,\gamma_h). \end{array}\right.$$

The computation begins from the indices $\{t+h-1, 1\}$, using $A_{t+h,0} = 0$ and $B_{t+h,0} = 0$, and progresses backward in time until reaching $\{t, h\}$.

Proof The result is clearly established for $h = 0$. Assume it is satisfied for any date t and any horizon $i \in \{1, \dots, h-1\}$, for $h \geq 0$. We have:

$$
\begin{aligned}
& \mathbb{E}_t[\exp(\gamma_1' w_{t+1} + \cdots + \gamma_h' w_{t+h}] \\
= \quad & \mathbb{E}_t[\mathbb{E}_{t+1}[\exp(\gamma_1' w_{t+1} + \cdots + \gamma_h' w_{t+h})]] \\
= \quad & \mathbb{E}_t[\exp(\gamma_1' w_{t+1} + A_{t+1,h-1}(\gamma_2, \dots, \gamma_h)' w_{t+h} + B_{t+1,h-1}(\gamma_2, \dots, \gamma_h))],
\end{aligned}
$$

which leads to the result using (I.33).

Example I.26 (Nominal interest rates)

- *If $B(t,h)$ denotes the date-t price of a nominal zero-coupon bon of maturity h, then*

$$B(t,h) = \mathbb{E}_t^{\mathbb{Q}} exp(-r_t - \cdots - r_{t+h-1}), \tag{I.35}$$

where r_t is the nominal short rate between t and $t+1$ (observed at t), and the associated (continuously-compounded) yield-to-maturity is given by:

$$R(t,h) = -\frac{1}{h} \log B(t,h), \quad h = 1, \ldots, H. \tag{I.36}$$

- *If $r_t = \omega' w_t$ (say), then:*

$$B(t,h) = \exp(-r_t)\mathbb{E}_t^{\mathbb{Q}} \exp(-\omega' w_{t+1} - \cdots - \omega' w_{t+h-1})$$

$$\Rightarrow u_1 = 0, \text{ and } u_i = -\omega, i = 2, \ldots, h \text{ (in I.31)}.$$

- *$B(t,h)$ exponential affine in w_t, $R(t,h)$ affine in w_t.*

Example I.27 (Real interest rates)

- *Denoting by q_t the price index on date t and by $\pi_{t+1} = log \dfrac{q_{t+1}}{q_t}$ the inflation rate on date $t+1$, we have:*

$$
\begin{aligned}
\bar{R}(t,h) &= -\frac{1}{h} \log \bar{B}(t,h), \quad h = 1, \dots, H \\
\bar{B}(t,h) &= \mathbb{E}_t^{\mathbb{Q}} \exp(-r_t - \dots - r_{t+h-1} + \pi_{t+1} + \dots + \pi_{t+h}), \\
&= \exp(-r_t) \times \\
&\quad \mathbb{E}_t^{\mathbb{Q}} \exp(-r_{t+1} - \dots - r_{t+h-1} + \pi_{t+1} + \dots + \pi_{t+h})
\end{aligned}
$$

- *If $r_t = \omega' w_t$ and $\pi_t = \bar{\omega}' w_t$ then $\bar{B}(t,h)$ is given by:*

$$\exp(-r_t)\mathbb{E}_t^{\mathbb{Q}} exp[(\bar{\omega} - \omega)' w_{t+1} + \dots + (\bar{\omega} - \omega)' w_{t+h-1} + \bar{\omega}' w_{t+h}]$$

$\Rightarrow u_1 = \bar{\omega}$ *and* $u_i = \bar{\omega} - \omega$, $i = 2, \dots, h$ *(in I.31).*

Example I.28 (Futures)

- $F(t,h) = \mathbb{E}_t^{\mathbb{Q}}(S_{t+h})$, $h = 1, \dots, H$, *where* S_t *date-t price of the asset.*
- *If* $w_t = (\log S_t, x_t')'$, *with* x_t *a vector of additional factors, then*

$$F(t,h) = \mathbb{E}_t^{\mathbb{Q}} \exp(e_1' w_{t+h}) \Rightarrow u_1 = e_1, \text{ and } u_i = 0,\ i = 2, \dots, h.$$

- *If* $w_t = (y_t, x_t')'$ *with* $y_t = \log \frac{S_t}{S_{t-1}}$, *then:*

$$F(t,h) = S_t \mathbb{E}_t^{\mathbb{Q}} \exp(e_1' w_{t+1} + \cdots + e_1' w_{t+h}) \Rightarrow u_i = e_1',\ i = 1, \dots, h.$$

Example I.29 (Exponential affine payoffs)

- *Consider, for* $h \in \{1, \dots, H\}$, *the date-t price of the payoff* $\exp(\nu' w_{t+h})$, *settled on date h:*

$$P(t,h;\nu) = \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \cdots - r_{t+h-1}) \exp(\nu' w_{t+h})].$$

- *If* $r_t = \omega' w_t$, *we get:*

$$P(t,h;\nu) = \exp(-r_t) \mathbb{E}_t^{\mathbb{Q}} \left(exp[-\omega' w_{t+1} - \cdots - \omega' w_{t+h-1} + \nu' w_{t+h}] \right)$$

$\Rightarrow u_1 = \nu$ *and* $u_i = -\omega$, $i = 2, \dots, h$ *(in* I.31*).*

Example I.29 (Exponential affine payoff (cont'd))

- *What precedes can be extended to the case where the payoff (settled on date $t+h$) is of the form:*

$$(\nu_1' w_{t+h}) \exp(\nu_2' w_{t+h}).$$

- *Indeed, we have*

$$\left[\frac{\partial \exp[(s\nu_1 + \nu_2)' w_{t+h}]}{\partial s}\right]_{s=0} = (\nu_1' w_{t+h}) \exp(\nu_2' w_{t+h}).$$

- *Therefore:*

$$\begin{aligned} & \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \cdots - r_{t+h-1})(\nu_1' w_{t+h}) \exp(\nu_2' w_{t+h})] \\ = & \left[\frac{\partial P(t, h; s\nu_1 + \nu_2)}{\partial s}\right]_{s=0}. \end{aligned} \quad \text{(I.37)}$$

- *This method is easily extended to price payoffs of the form $(\nu_1' w_{t+h})^k \exp(\nu_2' w_{t+h})$, $k \in \mathbb{N}$.*

I. Second Key Property: VAR representation

Prop. I.3 (VAR representation of an affine process' dynamics)

If w_t is the affine process whose Laplace transform is defined in Def. I.1, then its dynamics admits the following vectorial autoregressive representation:

$$w_{t+1} = \mu + \Phi w_t + \Sigma^{\frac{1}{2}}(w_t)\varepsilon_{t+1}, \tag{I.38}$$

where ε_{t+1} is a difference of martingale sequence whose conditional covariance matrix is the identity matrix and where μ, Φ and $\Sigma(w_t) = \Sigma^{\frac{1}{2}}(w_t)\Sigma^{\frac{1}{2}}(w_t)'$ satisfy:

$$\mu = \left[\frac{\partial}{\partial u} b(u)\right]_{u=0}, \quad \Phi = \left[\frac{\partial}{\partial u} a(u)'\right]_{u=0} \tag{I.39}$$

$$\Sigma(w_t) = \left[\frac{\partial}{\partial u \partial u'} b(u)\right]_{u=0} + \left[\frac{\partial}{\partial u \partial u'} a(u)' w_t\right]_{u=0}. \tag{I.40}$$

Proof. When w_t is affine, its (conditional) cumulant generating function is of the form $\psi(u) = a(u)'w_t + b(u)$. The result directly follows from the formulas given in Section I.3 (see top of Slide 11). ■

Remark I.2 (Multi-horizon conditional means and variances)

Proposition I.3 implies that:

$$\mathbb{E}_t(w_{t+h}) = (I-\Phi)^{-1}(I-\Phi^h)\mu + \Phi^h w_t \quad (I.41)$$

$$(I.42)$$

$$\mathbb{V}ar_t(w_{t+h}) = \Sigma(\mathbb{E}_t(w_{t+h-1})) + \Phi\Sigma(\mathbb{E}_t(w_{t+h-2}))\Phi' + \dots + \Phi^{h-1}\Sigma(w_t)\Phi^{h-1\prime}. \quad (I.43)$$

Eq. (I.43) notably shows that $\mathbb{V}ar_t(w_{t+h})$ is an affine function of w_t (indeed $\Sigma(.)$ is an affine function and the conditional expectations $\mathbb{E}_t(w_{t+h})$ are affine in w_t, as shown in Eq. I.41). We also have:

$$\left\{\begin{array}{rcl} \mathbb{E}(w_t) & = & (I-\Phi)^{-1}\mu \\ vec[\mathbb{V}ar(w_t)] & = & (I_{n^2} - \Phi\otimes\Phi)^{-1} vec\left(\Sigma[(I-\Phi)^{-1}\mu]\right). \end{array}\right. \quad (I.44)$$

Proof. See next slide.

Proof of Remark I.2. Eq. (I.41) easily deduced from Eq. (I.38), using that $\mathbb{E}_t(\varepsilon_{t+k})=0$ for $k>0$. As regards Eq. (I.43):

$$\mathbb{V}ar_t(w_{t+h}) \quad = \quad \mathbb{V}ar_t\left(\Sigma(w_{t+h-1})^{\frac{1}{2}}\varepsilon_{t+h}+\cdots+\Phi^{h-1}\Sigma(w_t)^{\frac{1}{2}}\varepsilon_{t+1}\right).$$

The conditional expectation at t of all the terms of the sum is equal to zero since, for $i\geq 1$:

$$\mathbb{E}_t\left[\Sigma(w_{t+i-1})^{\frac{1}{2}}\varepsilon_{t+i}\right]=\mathbb{E}_t[\underbrace{\mathbb{E}_{t+i-1}\{\Sigma(w_{t+i-1})^{\frac{1}{2}}\varepsilon_{t+i}\}}_{=\Sigma(w_{t+i-1})^{\frac{1}{2}}\mathbb{E}_{t+i-1}\{\varepsilon_{t+i}\}=0}\}],$$

and $\forall i<j$,

$$\mathbb{C}ov_t\left[\Sigma(w_{t+i-1})^{\frac{1}{2}}\varepsilon_{t+i},\Sigma(w_{t+j-1})^{\frac{1}{2}}\varepsilon_{t+j}\right]=\mathbb{E}_t\left[\Sigma(w_{t+i-1})^{\frac{1}{2}}\varepsilon_{t+i}\varepsilon'_{t+j}\Sigma'(w_{t+j-1})^{\frac{1}{2}}\right],$$

which can be seen to be equal to zero by conditioning on the information available on date $t+j-1$. Using the same conditioning, we obtain that:

$$\begin{aligned}
\mathbb{V}ar_t\left[\Phi^{h-j}\Sigma(w_{t+j-1})^{\frac{1}{2}}\varepsilon_{t+j}\right] &= \mathbb{E}_t\left[\Phi^{h-j}\Sigma(w_{t+j-1})^{\frac{1}{2}}\varepsilon_{t+j}\varepsilon'_{t+j}\Sigma'(w_{t+j-1})^{\frac{1}{2}}\Phi^{h-j\prime}\right]\\
&= \mathbb{E}_t\left[\Phi^{h-j}\Sigma(w_{t+j-1})^{\frac{1}{2}}\mathbb{E}_{t+j-1}(\varepsilon_{t+j}\varepsilon'_{t+j})\Sigma'(w_{t+j-1})^{\frac{1}{2}}\Phi^{h-j\prime}\right]\\
&= \Phi^{h-j}\mathbb{E}_t[\Sigma(w_{t+j-1})]\Phi^{h-j\prime}\\
&= \Phi^{h-j}\Sigma(\mathbb{E}_t[w_{t+j-1}])\Phi^{h-j\prime},
\end{aligned}$$

where the last equality results from the fact that he fact that $\Sigma(.)$ is affine (see Eq. I.40). ■

Definition I.4

A process $\{w_{1,t}\}$ is said to be Extended Affine or Extended Car (of order 1) if there exists a process $\{w_{2,t}\}$ such that $w_t = (w'_{1,t}, w'_{2,t})'$ is affine. Moreover:

- *if the σ-algebra $\sigma(\underline{w}_{1t})$ spanned by $\underline{w}_{1t}$ is equal to $\sigma(\underline{w}_t)$, $\{w_{1,t}\}$ will be called Internally Extended affine process.*
- *Otherwise, if $\sigma(\underline{w}_{1t}) \subset \sigma(\underline{w}_t)$, $\{w_{1,t}\}$ will be called Externally Extended affine process.*

For instance:

- If the process $w_{1,t}$ is Car(p), then the process $w_t = (w'_{1,t}, w'_{2,t})'$, with $w_{2,t} = (w'_{1,t-1}, \ldots, w'_{1,t-p+1})'$, is Car(1) [see Darolles, Gourieroux and Jasiak (2006)], and $\sigma(\underline{w}_{1t}) = \sigma(\underline{w}_t)$.
- $w_{1,t}$ may be non-Markovian.

Example I.30 (Gaussian ARMA process)

- *Consider an* $ARMA(1,1)$ *process*

$$w_{1,t} - \varphi w_{1,t-1} = \varepsilon_t - \theta \varepsilon_{t-1}$$

$$|\varphi| < 1,\ |\theta| < 1,\ \varepsilon_t \sim i.i.d.\mathcal{N}(0, \sigma^2).$$

- $w_{1,t}$ *is not Markovian.*
- *Take* $w_{2,t} = \varepsilon_t = (1-\theta L)^{-1}(1-\varphi L)w_{1,t}$.
  We have:

$$\begin{pmatrix} w_{1,t+1} \\ w_{2,t+1} \end{pmatrix} = \begin{pmatrix} \varphi & -\theta \\ 0 & 0 \end{pmatrix} \begin{pmatrix} w_{1,t} \\ w_{2,t} \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \varepsilon_{t+1}$$

- *Hence* $(w_{1,t}, w_{2,t})'$ *is Gaussian* $VAR(1)$ *and, therefore, affine (of order 1). Clearly,* $\sigma(\underline{w}_{1t}) = \sigma(\underline{w}_t)$, *so* $\{w_{1,t}\}$ *is an Internally* $ECar(1)$.
- *Easily generalized to* $ARMA(p,q)$ *and* $VARMA(p,q)$.

Example I.31 (GARCH type process)

- *$w_{1,t}$ is defined by:*

$$w_{1,t+1} = \mu + \varphi w_{1,t} + \sigma_{t+1}\varepsilon_{t+1},$$

where $|\varphi| < 1$ and $\varepsilon_t \sim i.i.d.\mathcal{N}(0,1)$, and

$$\sigma_{t+1}^2 = \omega + \alpha\varepsilon_t^2 + \beta\sigma_t^2,$$

where $0 < \beta < 1$.

- *Consider $w_{2,t} = \sigma_{t+1}^2$ non-linear function of $\underline{w_{1,t}}$.*

- *It can be shown (see next slide) that:*

$$\begin{aligned}
&\mathbb{E}\left[\exp(u_1 w_{1,t+1} + u_2 w_{2,t+1}) | \underline{w_{1,t}}\right] \\
= \quad &\exp\left[u_1\mu + u_2\omega - \frac{1}{2}\log(1 - 2u_2\alpha)\right. \\
&\left. + u_1\varphi w_{1,t} + (u_2\beta + \frac{u_1^2}{2(1-2u_2\alpha)}) w_{2,t}\right],
\end{aligned}$$

which is exponential affine in $(w_{1,t}, w_{2,t})$ and Internally ECar(1).

Proposition: Affine property of the GARCH-type process

The process $w_t = (w_{1,t}, w_{2,t})'$ defined by:

$$\begin{cases} w_{1,t+1} & = & \mu + \varphi w_{1,t} + \sigma_{t+1}\varepsilon_{t+1} \quad |\,\varphi\,| < 1 \\ \sigma^2_{t+1} & = & \omega + \alpha\varepsilon^2_t + \beta\sigma^2_t \quad 0 < \beta < 1, \alpha > 0, \omega > 0 \\ w_{2,t} & = & \sigma^2_{t+1}, \quad \varepsilon_t \sim i.i.d.\mathcal{N}(0,1) \end{cases}$$

is affine.

Proof. Note that $w_{2,t}$ is function of $\underline{w_{1,t}}$

$$\begin{aligned} & \mathbb{E}[\exp(u_1 w_{1,t+1} + u_2 w_{2,t+1})|\underline{w_{1,t}}] \\ = \quad & \exp(u_1\mu + u_1\varphi w_{1,t} + u_2\omega + u_2\beta w_{2,t})\mathbb{E}[\exp(u_1\sigma_{t+1}\varepsilon_{t+1} + u_2\alpha\varepsilon^2_{t+1})|\underline{w_{1,t}}] \end{aligned}$$

and, using Lemma 4:

$$\begin{aligned} & \mathbb{E}[\exp(u_1 w_{1,t+1} + u_2 w_{2,t+1})|\underline{w_{1,t}}] \\ = \quad & \exp(u_1\mu + u_1\varphi w_{1,t} + u_2\omega + u_2\beta w_{2t})\exp\left[-\frac{1}{2}\log(1-2u_2\alpha) + \frac{u_1^2 w_{2,t}}{2(1-2u_2\alpha)}\right] \\ = \quad & \exp\left[u_1\mu + u_2\omega - \frac{1}{2}\log(1-2u_2\alpha) + u_1\varphi w_{1,t} + \left(u_2\beta + \frac{u_1^2}{2(1-2u_2\alpha)}\right)w_{2,t}\right], \end{aligned}$$

which is exponential affine in $(w_{1,t}, w_{2,t})$.

Example I.32 (Quadratic transformation of Gaussian AR(1) processes)

- *Let us consider the following Gaussian AR(1) process:*

$$x_{t+1} = \mu + \rho x_t + \varepsilon_{t+1}\,, \quad \varepsilon_{t+1} \sim IIN(0, \sigma^2)\,.$$

- *If $\mu = 0$ the process $w_{1,t} = x_t^2$ is Car(1).*

- *If $\mu \neq 0$, the process $w_{1,t} = x_t^2$ is not Car, however $w_t = (w_{1,t}, x_t)' = (w_{1,t}, w_{2,t})'$ is Car(1) [see Gourieroux and Sufana (2011)] and, thus, $w_{1,t}$ is ECar(1).*

- *We have that $\sigma(\underline{w}_{1t}) \subset \sigma(\underline{w}_t)$ and therefore $w_{1,t}$ is Externally ECar(1) process.*

Example I.33 (Switching regimes Gaussian AR(1) processes)

- *Let us consider a 2-state switching regimes Gaussian AR(1) process $\{w_{1,t}\}$ given by:*

$$w_{1,t+1} = \mu' w_{2,t+1} + \rho w_{1,t} + (\sigma' w_{2,t+1})\varepsilon_{t+1} \, ,$$

where $\varepsilon_{t+1} \sim IIN(0,1)$, $\mu' = (\mu_1, \mu_2)$, $\sigma' = (\sigma_1, \sigma_2)$, and where $\{w_{2,t}\}$ is a 2-state homogeneous Markov chain with $\pi(e_1, e_1) = p$ and $\pi(e_2, e_2) = q$, independent of $\{\varepsilon_t\}$.

- *The Laplace transform of $w_{1,t+1}$, conditionally to $\underline{w}_{1,t}$, is not exponential-affine.*

- *Nevertheless, the bivariate process $w_t = (w_{1,t}, w_{2,t})'$ is Car(1) [see Monfort and Pegoraro (2007)].*

- *$w_{1,t}$ is an Externally ECar(1) process, given the additional information introduced by the Markov chain.*

Example I.34 (Stochastic volatility in mean processes)

- *We can specify also a stochastic volatility in mean AR(1) process defined by:*

$$w_{1,t+1} = \mu_1 + \mu_2 w_{2,t+1} + \rho w_{1,t} + w_{2,t+1}^{1/2} \varepsilon_{t+1} ,$$

where $\varepsilon_{t+1} \sim IIN(0,1)$, and where $\{w_{2,t}\}$ is an ARG(1) process independent of $\{\varepsilon_t\}$. The process $\{w_{1,t}\}$ is an Externally ECar(1) since $w_t = (w_{1,t}, w_{2,t})$ is Car(1) and $\sigma(\underline{w}_{1t}) \subset \sigma(\underline{w}_t)$.

- *We can also consider a n-variate stochastic volatility in mean VAR(1) process defined by:*

$$w_{1,t+1} = \mu + Rw_{1,t} + \begin{bmatrix} Tr(S_1 w_{2,t+1}) \\ \vdots \\ Tr(S_n w_{2,t+1}) \end{bmatrix} + w_{2,t+1}^{1/2} \varepsilon_{t+1} ,$$

where $\varepsilon_{t+1} \sim IIN(0, I)$, of size n, R is a $(n \times n)$ matrix, the S_i's are $(n \times n)$ symmetric matrices (of risk premia), and $\{w_{2,t}\}$ is an n-dimensional Wishart Autoregressive process independent of $\{\varepsilon_t\}$.

- *Here, $\{w_{1,t}\}$ is an n-dimensional ECar(1) process, because $w_t = (w_{1,t}', vech(w_{2,t})')'$ is Car(1) [see Gourieroux, Jasiak and Sufana (2010)].*

I.6 – TRUNCATED LAPLACE TRANSFORMS OF AFFINE PROCESSES

- As mentioned previously, if we know the conditional Laplace transform, we can characterize the conditional distribution of w_t.
- Let us see how to derive the conditional PDF of linear combinations of w_t from the Laplace transform.
- This derivation follows from the calculation of the truncated Laplace transform. This object has relevant applications, in particular, in the domain of option pricing [see Duffie, Pan, Singleton (2000)].
- The truncated Laplace transform is defined as follows:

$$\tilde{\varphi}_t(u; v, \gamma) = \mathbb{E}\left[\exp(u' w_{t+1}) \mathbb{1}_{\{v' w_{t+1} < \gamma\}} \,\middle|\, \underline{w_t}\right], \; u, v \in \mathbb{R}^n, \gamma \in \mathbb{R}.$$

- The following proposition shows how the truncated Laplace transform relates to the untruncated one.

I.6 – TRUNCATED LAPLACE TRANSFORMS OF AFFINE PROCESSES

Prop. I.4 (Computation of truncated conditional expectation)

If the complex untruncated conditional Laplace transform of w_t is given by

$$\varphi_t(z) = \mathbb{E}\left[\exp(z' w_{t+1}) \,\middle|\, \underline{w_t}\right], \quad z \in \mathbb{C}^n,$$

where n is the dimension of vector w_t, we have:

$$\begin{aligned} &\mathbb{E}\left[\exp(u' w_{t+1}) \mathbb{1}_{\{v' w_{t+1} < \gamma\}} \,\middle|\, \underline{w_t}\right] \\ = \quad & \frac{\varphi_t(u)}{2} - \frac{1}{\pi} \int_0^\infty \frac{Im[\varphi_t(u+ivx)\exp(-i\gamma x)]}{x} dx, \end{aligned} \tag{I.45}$$

where Im stands for imaginary part.

- Note that the integral in (I.45) is one-dimensional, regardless of the dimensionality of w_t. Hence, in practice, it can be effectively approximated using a Riemann sum.
- To evaluate the conditional cumulative distribution function (CDF) of a specific component of w_t (the i^{th} component, say), it suffices to apply (I.45) with $u = 0$ and $v = e_i$, where e_i is the i^{th} column of the identity matrix of dimension n.

I.6 – TRUNCATED LAPLACE TRANSFORMS OF AFFINE PROCESSES

- In the context of affine processes, simple extensions of (I.45) will enable us to derive the truncated Laplace transform for linear combinations of $w_{t+1,T} := \{w_{t+1}, \dots, w_T\}$, which will be crucial for pricing long-term derivatives, for instance.

- In that case, we want to compute:

$$\tilde{\varphi}_t(u; v, \gamma) = \mathbb{E}_t[\exp(u' w_{t+1,T}) \mathbb{1}_{\{v' w_{t+1,T} < \gamma\}}].$$

- Consider the complex untruncated conditional Laplace transform:

$$\varphi_t(z) = \mathbb{E}_t[\exp(z' w_{t+1,T})], \quad z \in \mathbb{C}^{nT},$$

computed using the same recursive algorithm as in the real case.

- We have [see Duffie, Pan, Singleton (2000) and next slides]:

$$\tilde{\varphi}_t(u; v, \gamma) = \frac{\varphi_t(u)}{2} - \frac{1}{\pi} \int_0^{\infty} \frac{Im[\varphi_t(u + ivx) \exp(-i\gamma x)]}{x} dx. \tag{I.46}$$

where Im means imaginary part.

- Note that the integral in Eq. (I.46) is one dimensional (whatever the dimension of w_t).
- Application: Option pricing. The typical problem is to compute conditional expectation of the type (with $k > 0$):

$$\begin{aligned}
& \mathbb{E}_t \left([\exp(u_1' w_{t+1,T}) - k \, \exp(u_2' w_{t+1,T})]^+ \right) \\
= \; & \mathbb{E}_t \left([\exp(u_1' w_{t+1,T}) - k \, \exp(u_2' w_{t+1,T})] \mathbb{1}_{\{[\exp(u_1 - u_2)' w_{t+1,T}] > k\}} \right) \\
= \; & \tilde{\varphi}_t(u_1; u_2 - u_1, -\log \, k) - k \, \tilde{\varphi}_t(u_2; u_2 - u_1, -\log \, k).
\end{aligned}$$

Transform Analysis:

- We want to compute:

$$\tilde{\varphi}_t(u; v, \gamma) = \mathbb{E}_t[\exp(u'w)\mathbb{1}_{(v'w<\gamma)}]$$

(noting $w = \tilde{w}_{t+1,T}$) knowing the function $\varphi_t(z) = \mathbb{E}_t[\exp(z'w)]$, z with complex components. We omit the index t for notational simplicity.

- Let us first note that, for given u and v, $\tilde{\varphi}_t(u; v, \gamma)$ is a positive increasing bounded function of γ and therefore can be seen, when considered as a function of γ, as the c.d.f. of a positive finite measure on $\mathbb{R}$, the Fourier transform of which is:

$$\int_{\mathbb{R}} \exp(i\gamma x) d\tilde{\varphi}(u; v, \gamma) = \mathbb{E} \int_{\mathbb{R}} \exp(i\gamma x) d\tilde{\varphi}_w(u; v, \gamma),$$

where, for given $w, \tilde{\varphi}_w(u; v, \gamma)$ is the c.d.f. of the mass point $\exp(u'w)$ at $v'w$, so we get the function of x:

$$\begin{aligned} \int_{\mathbb{R}} \exp(i\gamma x) d\tilde{\varphi}(u; v, \gamma) &= \mathbb{E}[\exp(ixv'w)exp(u'w)] \\ &= \mathbb{E}[\exp(u + ivx)'w] \\ &= \varphi(u + ivx). \end{aligned}$$

Transform Analysis (cont'd):

- Let us now compute for any real number λ:

$$
\begin{aligned}
& A(x_0, \lambda) \\
= \; & \frac{1}{2\pi} \int_{-x_0}^{x_0} \frac{\exp(i\lambda x)\varphi(u - ivx) - \exp(-i\lambda)\varphi(u + ivx)}{ix} dx \\
= \; & \frac{1}{2\pi} \int_{-x_0}^{x_0} \left[\int_{\mathbb{R}} \frac{\exp[-ix(\gamma - \lambda)] - \exp[ix(\gamma - \lambda)]}{ix} d\tilde{\varphi}(u; v, \gamma) \right] dx \\
= \; & \frac{1}{2\pi} \int_{\mathbb{R}} \left[\int_{-x_0}^{x_0} \frac{\exp[-ix(\gamma - \lambda)] - \exp[ix(\gamma - \lambda)]}{ix} dx \right] d\tilde{\varphi}(u; v, \gamma)
\end{aligned}
$$

- Now :

$$
\begin{aligned}
& \frac{1}{2\pi} \int_{-x_o}^{x_o} \frac{\exp[-ix(\gamma - \lambda)] - \exp[ix(\gamma - \lambda)]}{ix} dx \\
& = \frac{-sign(\gamma - \lambda)}{\pi} \int_{-x_o}^{x_o} \frac{sin(x \mid \gamma - \lambda \mid)}{x} dx
\end{aligned}
$$

which tends to -sign$(\gamma - \lambda)$ when $x_o \to \infty$ (where sign$(\omega) = 1$ if $\omega > 0$, sign$(\omega) = 0$ if $\omega = 0$, sign$(\omega) = -1$ if $\omega < 0$).

Transform Analysis (cont'd):

- Therefore:

$$A(\infty,\lambda) = -\int_{\mathbb{R}} sign(\gamma-\lambda) d\tilde{\varphi}(u;v,\gamma) = -\mathbb{E}\int_{\mathbb{R}} sign(\gamma-\lambda) d\tilde{\varphi}_w(u;\theta,\gamma),$$

where $\tilde{\varphi}_w(u;v,\gamma)$ is the c.d.f. of the mass point $exp(u'w)$ at $v'w$. Furthermore:

$$\int_{\mathbb{R}} sign(\gamma-\lambda) d\tilde{\varphi}_w(u;v,\gamma) = \begin{cases} \exp(u'w) & if \quad \lambda < v'w \\ 0 & if \quad \lambda = v'w \\ -\exp(u'w) & if \quad \lambda > v'w \end{cases}$$

Therefore, we have, taking λ equal to the fixed value of γ appearing in (I.46):

$$\begin{aligned} A(\infty,\lambda) &= -\mathbb{E}[\exp(u'w)(1-\mathbb{1}_{(v'w<\lambda)}) - \exp(u'w)\mathbb{1}_{(v'w<\lambda)}] \\ &= -\varphi(u) + 2\tilde{\varphi}(u;v,\lambda) \end{aligned}$$

which gives:

$$\tilde{\varphi}(u;,v,\gamma) = \frac{\varphi(u)}{2} + \frac{1}{2}A(\infty,\gamma).$$

Transform Analysis (cont'd):

- Using the definition of function A, we get:

$$\begin{aligned}
\frac{1}{2}A(\infty,\gamma) &= \frac{1}{4\pi}\int_{-\infty}^{\infty}\frac{\exp(i\gamma x)\varphi(u-ivx)-\exp(-i\gamma x)\varphi(u+ivx)}{ix}dx \\
&= \frac{1}{2\pi}\int_{o}^{\infty}\frac{\exp(i\gamma x)\varphi(u-ivx)-\exp(-i\gamma x)\varphi(u+ivx)}{ix}dx \\
&= -\frac{1}{\pi}\int_{o}^{\infty}\frac{\mathcal{I}m[\exp(-i\gamma x)\varphi(u+ivx)]}{x}dx
\end{aligned}$$

and finally

$$\tilde{\varphi}(u;v,\gamma)=\frac{\varphi(u)}{2}-\frac{1}{\pi}\int_{o}^{\infty}\frac{\mathcal{I}m[\varphi(u+ivx)\exp(-i\gamma x)]}{x}dx.\blacksquare$$

Example I.35 (Exogenous short rate)

- *Consider an asset whose date-t price is p_t.*
- *One-period geometric stock return: $w_{1,t+1} = y_{t+1} = \log(p_{t+1}/p_t)$.*
- *Consider a European Call option written on this asset (strike K and maturing in $t+h$). Payoff: $(p_{t+1} - K)^+ = p_t(\exp(y_{t+1}) - \kappa)^+$, with $\kappa = K/p_t$. Deterministic (exogenous) risk-free rate r_t.*

$\Rightarrow$ *Option price:*

$$p_t \exp(-r_t - \cdots - r_{t+h-1})\mathbb{E}_t^{\mathbb{Q}}\left\{[\exp\ u_1' w_{t+1,t+h} - \kappa]^+\right\},$$

with $u_1 = e \otimes e_1$, where e is the h-dimensional vector with components 1 and e_1 is the 1^{st} column of I_n (y_t being the 1^{st} component of w_t, say).

Example I.36 (Endogenous short rate)

- *Stochastic (endogenous) short-term rate: $r_{t+1} = \omega_0 + \omega_1' w_t$.*
- *The price is:*

$$\begin{aligned} & p_t \mathbb{E}_t^{\mathbb{Q}}\left[\exp(-\omega_0 - \omega_1' w_t - \cdots - \omega_0 - \omega_1' w_{t+h-1})[\exp(u_1' w_{t+1,t+h}) - \kappa]^+\right] \\ = \quad & p_t \exp(-h\omega_0 - \omega_1' w_t)\mathbb{E}_t^{\mathbb{Q}}\left(\left[\exp(\tilde{u}_1' w_{t+1,t+h}) - \kappa\ \exp(u_2 w_{t+1,t+h})\right]^+\right), \end{aligned}$$

with $\tilde{u}_1' = u_1 + u_2$, [$u_1 = e \otimes e_1$ as before], and $u_2 = (-\omega_1', \ldots, -\omega_1', 0)'$.

Chapter II

PRICING AND RISK-NEUTRAL DYNAMICS

– Agenda –

II.1 – STOCHASTIC DISCOUNT FACTOR (SDF)

Absence of Arbitrage Approach

In this chapter, we introduce the notions of SDF, change of probability measures, and risk-neutral dynamics, along with a typology of asset pricing models in discrete-time (under AAO).

- ☐ The period of interest is $\mathcal{T} = \{0, 1, 2, ..., T^*\}$.
- The n-dimensional state vector w_t gives the new information in the economy at date t and the overall information at t is $\underline{w_t} = (w_t, w_{t-1}, \ldots, w_0)$.
- The historical, or physical, dynamics of w_t is defined by the joint distribution of $\underline{w_T}$ (denoted by $\mathbb{P}$) or, equivalently, by the conditional probability distributions $\mathbb{P}_{t,t+1}$ of w_{t+1} given $\underline{w_t}$, or by the conditional log-Laplace transform:

$$\psi_t(u \,|\, \underline{w_t}) = \log \varphi_t(u \,|\, \underline{w_t}) = \log \mathbb{E}[\exp(u' w_{t+1}) \,|\, \underline{w_t}].$$

- More formally, we consider the measurable space $(\mathcal{W}, \mathcal{F})$ where $\mathcal{W}$ is the set of possible values of $\underline{w_{T^*}}$ and $\mathcal{F}$ is the associated σ-algebra.

II.1 – STOCHASTIC DISCOUNT FACTOR (SDF)

Absence of Arbitrage Approach

- $\mathcal{W}$ is the product of the sets $\mathcal{W}_t$ of the possible values of w_t, $t = 0, \ldots, T^*$, and $\mathcal{F}$ is the product of the σ-algebras $\mathcal{F}_t$ associated with $\mathcal{W}_t$, $\mathcal{F}_t$ being in turn the product of the borelian σ-algebras corresponding to the continuous components of w_t and the sets of all subsets corresponding to discrete components.
- This measurable space becomes a fundamental probability space by adding either the joint probability distribution $\mathbb{P}$ or a conditional probability $\mathbb{P}_{t,\mathcal{T}}$ of $\underline{w_{T^*}}$ given $\underline{w_t}$, depending on the problem of interest.
- Let us denote by $L_{2t}, t \in \mathcal{T}$, the (Hilbert) space of square integrable functions[1] $g(\underline{w_t})$, defined in the probability space $(\mathcal{W}, \mathcal{F}, \mathbb{P})$. We have $L_{2t} \subset L_{2s}$ if $t < s$, and we make the following assumptions:

Assumption II.1 (Price existence and uniqueness)

For any $\underline{w_t}$ and for any $t < s$, there exists a unique $p_t[g(\underline{w_s})]$, function of $\underline{w_t}$, price at t of a payoff $g(\underline{w_s}) \in L_{2s}$ delivered at s.

Assumption II.2 (Linearity and continuity)

For any $t < s$, and for any $\underline{w_t}$, $g_1(\underline{w_s})$ and $g_2(\underline{w_s})$ in L_{2s}, we have:

- $p_t[\lambda_1 \, g_1(\underline{w_s}) + \lambda_2 \, g_2(\underline{w_s})] = \lambda_1 \, p_t[g_1(\underline{w_s})] + \lambda_2 \, p_t[g_2(\underline{w_s})]$,
- *If* $g_n(\underline{w_s}) \xrightarrow[n\to\infty]{L_{2s}} 0$, *then* $p_t[g_n(\underline{w_s})] \underset{n\to\infty}{\longrightarrow} 0$.

Prop. II.1 (Stochastic discount factor)

Under Assumptions II.1 and II.2, for all $\underline{w_t}$ and for all $t < s$, there exists a unique $\mathcal{M}_{t,s}(\underline{w_s}) \in L_{2s}$ such that, $\forall g(\underline{w_s}) \in L_{2s}$,

$$p_t[g(\underline{w_s})] = \mathbb{E}[\mathcal{M}_{t,s}(\underline{w_s}) \, g(\underline{w_s}) \mid \underline{w_t}]. \tag{II.1}$$

$\mathcal{M}_{t,s} = \mathcal{M}_{t,s}(\underline{w_s})$ is called the stochastic discount factor (SDF) over the time interval $[t, s]$. In particular, the price at the date t of a zero coupon bond paying $g(\underline{w_s}) = 1$ at the maturity date s is $\mathbb{E}(\mathcal{M}_{t,s} \mid \underline{w_t})$.

The previous Proposition II.1:

- does not imply anything about the sign of the SDF.
- In what follows, we establish that the absence of arbitrage opportunity (AAO) assumption is equivalent to a strictly positive SDF or,
- equivalently, to a pricing function p_t giving strictly positive prices to non-negative payoffs that are strictly positive with some positive probability (conditioned on $\underline{w_t}$).

Assumption II.3 (Absence of Arbitrage Opportunity (AAO))

At any date t, it is impossible to constitute a portfolio of future payoffs, possibly modified at subsequent dates, such that:

- *the price of the portfolio at t is non-positive,*
- *the payoffs at subsequent dates are non-negative, and*
- *there is at least one subsequent date s such that the net payoff at s is strictly positive with a non zero conditional probability at t.*

Prop. II.2 (Strictly positive $\mathcal{M}_{t,s}$)

Suppose Assumptions II.1 and II.2 are satisfied. Then, Assumption II.3 is satisfied $\Leftrightarrow \mathbb{P}(\mathcal{M}_{t,s} > 0 \,|\, \underline{w_t}) = 1$, *for all* $\underline{w_t}$ *and for all* $s > t$.

Proof. $\Leftarrow$ is obvious. If $\Rightarrow$ was not true, the payoff $\mathbb{1}_{\{\mathcal{M}_{t,s} \leq 0\}}$, at s, would be such that $\mathbb{P}[\mathbb{1}_{\{\mathcal{M}_{t,s} \leq 0\}} = 1 | \underline{w_t}] > 0$ and $p_t[\mathbb{1}_{\{\mathcal{M}_{t,s} \leq 0\}}] = \mathbb{E}_t[\mathcal{M}_{t,s}\mathbb{1}_{\{\mathcal{M}_{t,s} \leq 0\}}] \leq 0$, thus contradicting the assumption of no arbitrage. ■

Prop. II.3

For any sequence of dates $t < r < s$, *for any* $\underline{w_t}$, *and for any* $g(\underline{w_s}) \in L_{2s}$, *the no-arbitrage pricing function* $p_t(\cdot)$ *is such that:*

$$p_t[g(\underline{w_s})] = p_t\{p_r[g(\underline{w_s})]\}. \quad \text{(II.2)}$$

Proof. If this was not true, we could construct over the time interval $[t, s]$ a sequence of portfolios with a strictly positive payoff at s, with zero payoffs at any date $r \in]t, s[$, and with price zero at t, contradicting Assumption II.3. Indeed, assuming, for instance, $p_t(g_s) > p_t[p_r(g_s)]$, the sequence of portfolios, and the payoff at s, are defined by the following trading strategy:

- at t: buy the payoff $p_r(g_s)$, (short) sell the payoff g_s, and buy $\dfrac{p_t(g_s) - p_t[p_r(g_s)]}{\mathbb{E}(\mathcal{M}_{t,s}|\underline{w_t})}$ units of zero-coupon bonds, paying a unitary payoff at s; the associated portfolio has a value equal to zero;
- at r: buy the payoff g_s and sell the asset $p_r(g_s)$, generating a net payoff equal to zero;
- at s, the net payoff is: $g_s - g_s + \dfrac{p_t(g_s) - p_t[p_r(g_s)]}{\mathbb{E}(\mathcal{M}_{t,s}|\underline{w_t})} > 0$.

Similarly, if $p_t(g_s) < p_t[p_r(g_s)]$, we can contradict Assumption II.3, which proves the proposition.

Prop. II.4 (Time consistency)

For all $t < k < s$ and for all $\underline{w_t}$, under Assumption II.3 we have $\mathcal{M}_{t,s} = \mathcal{M}_{t,k}\,\mathcal{M}_{k,s}$, which implies:

- $\mathcal{M}_{t,s} = \mathcal{M}_{t,t+1}\,\mathcal{M}_{t+1,t+2}\ldots\mathcal{M}_{s-1,s}$;
- $\mathcal{M}_{0,t} = \Pi_{j=0}^{t-1}\,\mathcal{M}_{j,j+1}$ *($\mathcal{M}_{0,t}$ is called pricing kernel).*

Proof. Using Proposition II.3, we have:

$$
\begin{aligned}
p_t(g_s) &= \mathbb{E}(\mathcal{M}_{t,s}\,g_s|\underline{w_t}) = \mathbb{E}(\mathcal{M}_{t,r}\,p_k(g_s)\,|\,\underline{w_t}) \\
&= \mathbb{E}[\mathcal{M}_{t,k}\mathbb{E}(\mathcal{M}_{k,s}\,g_s\,|\,\underline{w_k})\,|\,\underline{w_t}] = \mathbb{E}(\mathcal{M}_{t,k}\,\mathcal{M}_{k,s}\,g_s|\underline{w_t}), \forall g, \forall \underline{w}_t,
\end{aligned}
$$

and, therefore, $\mathcal{M}_{t,s} = \mathcal{M}_{t,k} \times \mathcal{M}_{k,s}$.

- Consider an asset whose payoff, at date s, is $g(\underline{w_s})$. We have, for all $t < s$:

$$
\begin{aligned}
p_t[g(\underline{w_s})] &= \mathbb{E}[\mathcal{M}_{t,t+1}\times\ldots\times\mathcal{M}_{s-1,s}\,g(\underline{w_s})\,|\,\underline{w_t}] \\
&= \mathbb{E}_t[\mathcal{M}_{t,t+1}\times\ldots\times\mathcal{M}_{s-1,s}\,g(\underline{w_s})].
\end{aligned}
\tag{II.3}
$$

- In particular, since $L_{2,t+1}$ contains $g(\underline{w_{t+1}}) = 1$, the price at date t of a zero-coupon bond (ZCB) with residual maturity one is given by:

$$B(t,1) = \mathbb{E}_t[\mathcal{M}_{t,t+1}]. \tag{II.4}$$

- If we denote by r_t the (predetermined) continuously compounded risk-free short rate, defined by $B(t,1) = \exp(-r_t)$, we get:

$$r_t = -\log \mathbb{E}_t[\mathcal{M}_{t,t+1}], \tag{II.5}$$

and if we denote by $B_{t,h}$ the price at date t of a ZCB with residual maturity h, we have:

$$B(t,h) = \mathbb{E}_t[\mathcal{M}_{t,t+1} \times \ldots \times \mathcal{M}_{t+h-1,t+h}] = \mathbb{E}_t[\mathcal{M}_{t,t+h}], \tag{II.6}$$

with the associated continuously compounded yield-to-maturity given by:

$$R(t,h) = -\frac{1}{h} \log \mathbb{E}_t[\mathcal{M}_{t,t+h}]. \tag{II.7}$$

Definition II.1 (Price process)

A (no-arbitrage) price process p_t*,* $t \in \mathcal{T}$*, is defined by* $p_t = \mathbb{E}_t[\mathcal{M}_{t,T^*} A_{T^*}]$*, where* $A_{T^*} \in L_{2T^*}$ *is a payoff function of* $\underline{w_{T^*}}$*.*

- A particular price process that will be used to define the risk-neutral probability measure is the bank or money-market account.

Definition II.2 (Bank or money-market account)

The bank (or money-market) account process $A_{0,t}$ *(function of* $\underline{w_{t-1}}$*) is defined by:*

$$A_{0,t} = \exp(r_0 + \ldots + r_{t-1}) = \frac{1}{\mathbb{E}_0[\mathcal{M}_{0,1}] \times \ldots \mathbb{E}_{t-1}[\mathcal{M}_{t-1,t}]}, \qquad \text{(II.8)}$$

and it is the price process of the payoff A_{0,T^*} *since*

$$A_{0,t} = \mathbb{E}_t[\mathcal{M}_{t,t+1} \times \ldots \times \mathcal{M}_{T^*-1,T^*} A_{0,T^*}].$$

- In other words, $A_{0,t}$ is the value of an investment started at date $t = 0$, when it was worth one unit of money, and invested on each date for one period at the risk-free short-rate r_t.

Prop. II.5 (Martingale property)

For all $t < s$, and for any $\underline{w_t}$, a no-arbitrage price process p_t is such that:

$$p_t = \mathbb{E}_t[\mathcal{M}_{t,s}\, p_s] \text{ or } \mathcal{M}_{0,t}\, p_t = \mathbb{E}_t[\mathcal{M}_{0,s}\, p_s], \tag{II.9}$$

that is, $(\mathcal{M}_{0,t}\, p_t)$ is a $\mathbb{P}$-martingale. In particular, $(\mathcal{M}_{0,t}\, A_{0,t})$ is a $\mathbb{P}$-martingale.

- Equation (II.9), also known as the Euler equation, is frequently used in structural models, particularly to derive a stochastic discount factor from specified agent preferences.
- Up to now, we have characterized the price p_t at the date t of a payoff $g(\underline{w_s})$ at date s as the conditional (at t) expectation (under $\mathbb{P}$) of the product between the payoff $g(\underline{w_s})$ and the SDF $\mathcal{M}_{t,s}$.
- In the next section we show that this pricing formula (see relation (II.9)) can be equivalently written w.r.t. another probability measure, equivalent to $\mathbb{P}$, called risk-neutral (RN) probability measure or equivalent martingale measure (EMM).

II.2 – STOCHASTIC DISCOUNT FACTOR (SDF)

Change of Probability Measure

Prop. II.6 (Numeraire and change of probability measure)

A numeraire is a price process (N_t) *with/* $N_0 = 1$*. Let* $\mathbb{Q}^N$ *the probability measure, equivalent to* $\mathbb{P}$*, defined by the following sequence of conditional (one-period Radon-Nikodym) densities, and some initial value* w_0*:*

$$\frac{d\mathbb{Q}^N_{t,t+1}}{d\mathbb{P}_{t,t+1}} = \frac{N_{t+1}\mathcal{M}_{t,t+1}}{N_t} > 0\,,\ \mathbb{E}_t\left(\frac{d\mathbb{Q}^N_{t,t+1}}{d\mathbb{P}_{t,t+1}}\right) = 1\,,\ t \in \{0,\dots,T^*-1\}.$$

$\dfrac{d\mathbb{Q}^N_{t,t+1}}{d\mathbb{P}_{t,t+1}}$ *is the density function of the one-period conditional probability of* w_{t+1} *given* $\underline{w_t}$ *under* $\mathbb{Q}^N$*, with respect to the same conditional probability under* $\mathbb{P}$*. The associated joint density of* $\mathbb{Q}^N$ *with respect to* $\mathbb{P}$ *is:*

$$\begin{aligned}\xi_{T^*} &= \frac{d\mathbb{Q}^N}{d\mathbb{P}} = \prod_{t=0}^{T^*-1} \frac{d\mathbb{Q}^N_{t,t+1}}{d\mathbb{P}^N_{t,t+1}} \\ &= \prod_{t=0}^{T^*-1} \frac{N_{t+1}\,\mathcal{M}_{t,t+1}}{N_t} = N_{T^*}\,\mathcal{M}_{0,T^*}.\end{aligned} \tag{II.10}$$

Prop. II.6 (Numeraire and change of probability measure)

Denoting ξ_t the density function of the joint probability of $\underline{w_t}$ under $\mathbb{Q}^N$, with respect to the same probability under $\mathbb{P}$, we have:

$$\xi_t = \prod_{j=0}^{t-1} \frac{d\mathbb{Q}^N_{j,j+1}}{d\mathbb{P}_{j,j+1}} = \mathbb{E}_t(\xi_{T^*}),$$

and, therefore, (ξ_t) is a $\mathbb{P}$-martingale.

Prop. II.7 (Numeraire and price process)

Let us consider a numeraire N_t and a (no-arbitrage) price process p_t. Then, the process $\dfrac{p_t}{N_t}$ is a $\mathbb{Q}^N$-martingale.

Proof From (II.9) we have:

$$p_t = \mathbb{E}_t(\mathcal{M}_{t,t+1}\, p_{t+1}) \Leftrightarrow \frac{p_t}{N_t} = \mathbb{E}_t\left(\frac{N_{t+1}\,\mathcal{M}_{t,t+1}}{N_t}\,\frac{p_{t+1}}{N_{t+1}}\right) = \mathbb{E}_t^{\mathbb{Q}^N}\left(\frac{p_{t+1}}{N_{t+1}}\right). \tag{II.11}$$

Prop. II.8 (The risk-neutral probability measure)

Let us consider as numeraire the bank account process $(A_{0,t})$. Then, the density function of the conditional probability of w_{t+1}, given $\underline{w_t}$, under $\mathbb{Q}$ with respect to the same conditional probability under $\mathbb{P}$ is given by:

$$\frac{d\mathbb{Q}_{t,t+1}}{d\mathbb{P}_{t,t+1}} = \frac{A_{0,t+1}\mathcal{M}_{t,t+1}}{A_{0,t}} = \frac{\mathcal{M}_{t,t+1}}{\mathbb{E}_t[\mathcal{M}_{t,t+1}]} = \exp(r_t)\,\mathcal{M}_{t,t+1} > 0, \tag{II.12}$$

and $\mathbb{Q}$ is called the risk-neutral (RN) probability measure. For any time interval $[t,s] \in \mathcal{T}$, the density function of the joint conditional distribution of $(w_{t+1},\dots,w_s)$, given $\underline{w_t}$, under $\mathbb{Q}$ with respect to the same distribution under $\mathbb{P}$ is:

$$\begin{aligned}\frac{d\mathbb{Q}_{t,s}}{d\mathbb{P}_{t,s}} &= \frac{\mathcal{M}_{t,t+1} \times \ldots \times \mathcal{M}_{s-1,s}}{\mathbb{E}_t[\mathcal{M}_{t,t+1}] \times \ldots \times \mathbb{E}_t[\mathcal{M}_{s-1,s}]} \\ &= \exp(r_t + \ldots + r_{s-1})\,\mathcal{M}_{t,s}.\end{aligned}$$

Prop. II.9 (Pricing with $\mathbb{Q}$)

Proposition II.7 implies that the discounted asset price $\left(\dfrac{p_t}{A_{0,t}}\right)$ is a $\mathbb{Q}$-martingale. The price at $t < s$ of a payoff $g(\underline{w_s})$ at s is given by:

$$p_t = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{A_{0,t}}{A_{0,s}}\, g(\underline{w_s})\right] = \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \ldots - r_{s-1})\, g(\underline{w_s})]. \tag{II.13}$$

In particular, the price at date t of the ZCB with residual maturity h can be written:

$$\begin{aligned} B(t,h) &= \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \ldots - r_{t+h-1})] \\ &= \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t)B(t+1,h-1)], \end{aligned} \tag{II.14}$$

and the yield-to-maturity can be written :

$$R(t,h) = -\frac{1}{h}\log \mathbb{E}_t^{\mathbb{Q}}\left[\exp(-r_t - \ldots - r_{t+h-1})\right]. \tag{II.15}$$

II.3 – THE EXPONENTIAL-AFFINE SDF

- We assume $\mathcal{M}_{t,t+1} = \mathcal{M}_{t,t+1}(\underline{w_{t+1}})$ has an exponential-affine form:

$$\mathcal{M}_{t,t+1}(\underline{w_{t+1}}) = \exp\left[\alpha_t(\underline{w_t})' w_{t+1} + \beta_t(\underline{w_t})\right],$$

  $\alpha_t(\underline{w_t})$ is the risk sensitivity vector, or vector of prices of risk.

- Given that:

$$\mathbb{E}_t[\mathcal{M}_{t,t+1}] = \exp[-r_t(\underline{w_t})] = \exp[\psi_t(\alpha_t(\underline{w_t})) + \beta_t(\underline{w_t})],$$

  $\psi_t(u)$ being the conditional log-LT of w_{t+1} given $\underline{w_t}$, we have:

$$\mathcal{M}_{t,t+1} = \exp\left[-r_t(\underline{w_t}) + \alpha_t'(\underline{w_t})\, w_{t+1} - \psi_t(\alpha_t)\right]. \qquad \text{(II.16)}$$

- The exponential-affine specification of the SDF, along with the affine nature of the risk-neutral dynamics of the state vector (w_t), will play a key role in obtaining explicit or quasi-explicit pricing formulas in our no-arbitrage framework.

II.3 – THE EXPONENTIAL-AFFINE SDF

- This specification can be justified from a structural perspective. In the Consumption-CAPM, the representative agent has a power utility function:

$$u(C_t) = (C_t^{1-\gamma} - 1)/(1-\gamma),$$

with $\gamma > 0$, and C_t being the consumption at t of the single good with price q_t. The maximization of the expected utility implies the following nominal stochastic discount factor:

$$\mathcal{M}_{t,t+1} = \exp\left[\log\delta - \log\left(\frac{q_{t+1}}{q_t}\right) - \gamma\log\left(\frac{C_{t+1}}{C_t}\right)\right],$$

where δ is the intertemporal discount rate.

- Hence, in that case, $\mathcal{M}_{t,t+1}$ is exponential-affine in $w_{t+1} = (\pi_{t+1}, \Delta c_{t+1})'$, where $\pi_{t+1} = \log(q_{t+1}/q_t)$ is the inflation rate between t and $t+1$, and $\Delta c_{t+1} = \log(C_{t+1}/C_t)$ is the (log) consumption growth rate.

II.3 – THE EXPONENTIAL-AFFINE SDF

- The vector of prices of risk $\alpha_t(\underline{w_t})$ is closely related to the maximum Sharpe ratio that prevails in the economy.

- The Sharpe ratio of a price process p_t is defined as the ratio between the expectation of its excess return and its standard deviation.
- Specifically, the Sharpe ratio of an asset of price p_t is given by:

$$\frac{\mathbb{E}_t(R_{t+1} - R_{f,t})}{\sigma_t(R_{t+1})},$$

where $R_{t+1} = (p_{t+1} - p_t)/p_t$ is the arithmetic return of this process, and $\sigma_t(R_{t+1})$ denotes the conditional standard deviation of this return; $R_{f,t}$ is the arithmetic short-term risk-free rate.

- Grossman and Shiller (1981) and Hansen and Jagannathan (1991) have shown that the Euler equation (II.9) implies an upper bound for this ratio.

II.3 – THE EXPONENTIAL-AFFINE SDF

- Since the price of a one period bond is given by $\mathbb{E}_t(\mathcal{M}_{t,t+1})$ and also by $1/(1+R_{f,t})$ (by definition of the short-term interest rate $R_{f,t}$), it comes that

$$\mathbb{E}_t(\mathcal{M}_{t,t+1}) = \frac{1}{(1+R_{f,t})}.$$

- Using the latter in the Euler equation $\mathbb{E}_t[\mathcal{M}_{t,t+1}(p_{t+1}/p_t)] = 1$, we get:

$$1 = \frac{\mathbb{E}_t(1+R_{t+1})}{(1+R_{f,t})} + \mathbb{C}ov_t\left(\mathcal{M}_{t,t+1}, R_{t+1}\right),$$

which leads to:

$$\mathbb{E}_t(R_{t+1} - R_{f,t}) = -\mathbb{C}ov_t\left(\frac{\mathcal{M}_{t,t+1}}{\mathbb{E}_t(\mathcal{M}_{t,t+1})}, R_{t+1}\right). \qquad \text{(II.17)}$$

II.3 – THE EXPONENTIAL-AFFINE SDF

- Applying the Cauchy-Schwarz inequality to the covariance term gives:

$$\underbrace{\left|\frac{\mathbb{E}_t(R_{t+1}-R_{f,t})}{\sigma_t(R_{t+1})}\right|}_{\substack{\text{Sharpe ratio of the}\\ \text{considered asset}}} \leq \underbrace{\frac{\sigma_t(\mathcal{M}_{t,t+1})}{\mathbb{E}_t(\mathcal{M}_{t,t+1})}}_{\substack{\text{Maximum Sharpe}\\ \text{ratio in the economy}}} . \qquad \text{(II.18)}$$

- When the SDF admits the exponential-affine representation (II.16), the maximum Sharpe ratio is a function of the vector of prices of risk:

$$\frac{\sigma_t(\mathcal{M}_{t,t+1})}{\mathbb{E}_t(\mathcal{M}_{t,t+1})} = \sqrt{\exp[\psi_t(2\alpha_t)-2\psi_t(\alpha_t)]-1}. \qquad \text{(II.19)}$$

- Indeed, the square of this ratio is $\mathbb{V}ar_t \exp[\alpha_t'(\underline{w_t})\, w_{t+1} - \psi_t(\alpha_t)] = \mathbb{E}_t \exp[2\alpha_t'(\underline{w_t})\, w_{t+1} - 2\psi_t(\alpha_t)] - 1$.

II.3 – THE EXPONENTIAL-AFFINE SDF

- The Cauchy-Schwarz inequality states that for any pair of random variables X and Y:

$$|\mathbb{C}ov(X,Y)| \leq \sigma(X)\sigma(Y).$$

- The equality holds when X and Y are perfectly correlated.
- In the present case, the latter implies that the maximum Sharpe ratio is achieved by assets whose returns are perfectly correlated with the SDF's innovations.

II.4 – THE RISK-NEUTRAL DYNAMICS

- Let us denote by $f(w_{t+1} \mid \underline{w_t})$, or $f_t(w_{t+1})$, the density function of $\mathbb{P}_{t,t+1}$ with respect to an appropriate dominating measure (typically the product of Lebesgue measures and counting measures).

- From Proposition II.8, we know that:

$$\frac{d\mathbb{Q}_{t,t+1}}{d\mathbb{P}_{t,t+1}} = \exp(r_t)\,\mathcal{M}_{t,t+1},$$

- Thus, the probability distributions are equivalent and $\mathbb{Q}_{t,t+1}$ has a density with respect to the same dominating measure as $\mathbb{P}_{t,t+1}$ given by:

$$f_t^{\mathbb{Q}}(w_{t+1}) = f_t(w_{t+1})\,\mathcal{M}_{t,t+1}(\underline{w_{t+1}})\,\exp[r_t(\underline{w_t})]. \quad \text{(II.20)}$$

- When the SDF takes the specification (II.16), the density of the conditional distribution of w_{t+1}, given $\underline{w_t}$, under $\mathbb{Q}$ with respect to the same distribution under $\mathbb{P}$ is:

$$
\begin{aligned}
d^{\mathbb{Q}}(w_{t+1} \mid \underline{w_t}) &= \frac{f_t^{\mathbb{Q}}(w_{t+1})}{f_t(w_{t+1})} = \mathcal{M}_{t,t+1}(\underline{w_{t+1}}) \exp[r_t(\underline{w_t})] \\
&= \exp[\alpha_t' w_{t+1} - \psi_t(\alpha_t)],
\end{aligned}
\tag{II.21}
$$

and $d^{\mathbb{Q}}(w_{t+1} \mid \underline{w_t})$ is also exponential-affine.

- In addition, we have:

$$
f_t^{\mathbb{Q}}(w_{t+1}) = f_t(w_{t+1}) \frac{\exp(\alpha_t' w_{t+1})}{\varphi_t(\alpha_t)}, \tag{II.22}
$$

so $f_t^{\mathbb{Q}}(w_{t+1})$ is the Esscher transform of $f_t(w_{t+1})$ evaluated at α_t.

Prop. II.10 (Risk-neutral conditional Laplace transform)

The risk-neutral conditional Laplace transform of w_{t+1} given $\underline{w_t}$, denoted $\varphi_t^{\mathbb{Q}}(u) = \varphi_t^{\mathbb{Q}}(u \mid \underline{w_t})$, is given by:

$$\begin{aligned}
\varphi_t^{\mathbb{Q}}(u) &= \mathbb{E}_t^{\mathbb{Q}}\left[\exp(u' w_{t+1})\right] \\
&= \mathbb{E}_t\left\{\exp[(u+\alpha_t)' w_{t+1} - \psi_t(\alpha_t)]\right\} \\
&= \exp[\psi_t(u+\alpha_t) - \psi_t(\alpha_t)] = \frac{\varphi_t(u+\alpha_t)}{\varphi_t(\alpha_t)}.
\end{aligned}$$

Hence, the risk-neutral conditional log-Laplace transform is given by:

$$\psi_t^{\mathbb{Q}}(u) = \psi_t(u+\alpha_t) - \psi_t(\alpha_t). \qquad \text{(II.23)}$$

- It is also immediately seen that:

$$\begin{aligned}
d_t(w_{t+1}) = \frac{d\mathbb{P}_{t,t+1}}{d\mathbb{Q}_{t,t+1}} = \frac{1}{d_t^{\mathbb{Q}}(w_{t+1})} &= \frac{\exp(-r_t)}{\mathcal{M}_{t,t+1}} \\
&= \exp[-\alpha_t' w_{t+1} + \psi_t(\alpha_t)].
\end{aligned}$$

- Now, if we take $u = -\alpha_t$ in (II.23) we obtain $\psi_t^{\mathbb{Q}}(-\alpha_t) = -\psi_t(\alpha_t)$ since $\psi_t(0) = 0$, and replacing u by $(u - \alpha_t)$, we obtain that the historical conditional log-Laplace transform can be written as:

$$\psi_t(u) = \psi_t^{\mathbb{Q}}(u - \alpha_t) - \psi_t^{\mathbb{Q}}(-\alpha_t),$$

- and we can equivalently write:

$$\begin{cases} d_t(w_{t+1}) & = & \exp[-\alpha_t' w_{t+1} - \psi_t^{\mathbb{Q}}(-\alpha_t)] \\ d_t^{\mathbb{Q}}(w_{t+1}) & = & \exp[\alpha_t' w_{t+1} + \psi_t^{\mathbb{Q}}(-\alpha_t)]. \end{cases}$$

- Two relevant results stand out from Proposition II.10:

Remark II.1 (Preserving the affine property)

If the state vector (w_t) is an affine process with conditional historical log-Laplace transform $\psi_t(u) = a(u)'w_t + b(u)$ (see Definition I.1) and if $\alpha_t(\underline{w_t})$ is constant, then $\psi_t^{\mathbb{Q}}(u)$ is also affine.

- If the $\mathbb{P}$-dynamics of (w_t) is described by a Gaussian VAR process, then α_t is allowed to be affine (in w_t) and the $\mathbb{Q}$-dynamics will be affine as well.
- If the $\mathbb{P}$-dynamics of (w_t) is given by a VARG process, then α_t has to be constant to preserve the VARG (thus, affine) nature under $\mathbb{Q}$.

Remark II.2 (Risk-free pricing)

When $\alpha_t = 0$, $\psi_t^{\mathbb{Q}}(u) = \psi_t(u)$ (since $\psi_t(0) = 0$), and thus historical and risk-neutral conditional (log-) Laplace transforms coincide. In other words, when the investor does not consider the state vector (w_t) a source of risk to be priced, $\mathcal{M}_{t,t+1}$ reduces to the (one-period) risk-free discount factor $\exp(-r_t)$ and $\mathbb{P} \equiv \mathbb{Q}$.

II.5 – TYPOLOGY OF ECONOMETRIC ASSET PRICING MODELS

Definition II.3 (Econometric Asset Pricing Model (EAPM))

An Econometric Asset Pricing Model (EAPM) is defined by the following functions:

- $r_t(\underline{w_t})$,
- $f(w_{t+1}|\underline{w_t}))$ *[or* $\psi_t(u)$*]*,
- $\mathcal{M}_{t,t+1}(\underline{w_{t+1}})$,
- $f^{\mathbb{Q}}(w_{t+1}|\underline{w_t})$ *[or* $\psi_t^{\mathbb{Q}}(u)$*]*.

☐ The previous functions have to to be specified and parameterized. They are linked by:

$$f^{\mathbb{Q}}(w_{t+1}|\underline{w_t}) = f(w_{t+1}|\underline{w_t})\mathcal{M}_{t,t+1}(\underline{w_{t+1}})\exp[r_t(\underline{w_t}))].$$

- If $\mathcal{M}_{t,t+1}$ is exponential-affine:

$$\mathcal{M}_{t,t+1}(\underline{w_{t+1}}) = \exp\{-r_t(\underline{w_t}) + \alpha_t'(\underline{w_t})\, w_{t+1} - \psi_t[\alpha_t(w_t)]\}.$$

- Once $r_t(\underline{w_t})$ is specified we have to specify $\psi_t, \alpha_t, \psi_t^{\mathbb{Q}}$ linked by

$$\psi_t^{\mathbb{Q}}(u) = \psi_t(u+\alpha_t) - \psi_t(\alpha_t).$$

  We have 3 ingredients (of the EAPM) linked by one (no-arbitrage based) equation: two of them only can be chosen.

- We are going to present three ways of specifying an EAPM:
  1) the Direct Modelling (from $\mathbb{P}$ and α_t to $\mathbb{Q}$),
  2) the R.N. Constrained Direct Modelling (or Mixed Modelling),
  3) the Back Modelling (from $\mathbb{Q}$ and α_t to $\mathbb{P}$).

  In all approaches, we have to specify the status of the short rate r_t (developed in next slide).

- Purpose: being $\mathbb{Q}$-affine (at least) in order to have tractable pricing formulas.

Remark II.3 (The status of the short rate)

The short rate r_t is a function of $\underline{w_t}$, this function may be known or unknown by the econometrician:

i) *It is known in two cases:*

   a) *r_t is exogenous $\Rightarrow r_t(\underline{w_t})$ does not depend on $\underline{w_t}$,*

   b) *r_t is an endogenous factor $\Rightarrow r_t$ is a component of w_t.*

ii) *If the function $r_t(\underline{w_t})$ is unknown, it has to be specified parametrically:*

$$\left\{ r_t(\underline{w_t}, \tilde{\theta}), \tilde{\theta} \in \tilde{\Theta} \right\},$$

where $r_t(.,.)$ is a known function.

1) The Direct Modelling

- **Step 1: Specification of the historical dynamics**
  We choose a parametric family for the conditional historical Log-Laplace transform $\psi_t(u \,|\, \underline{w_t})$:

$$\left\{\psi_t(u|\underline{w_t};\theta_{\mathbb{P}}), \theta_{\mathbb{P}} \in \Theta_{\mathbb{P}}\right\}.$$

- **Step 2: Specification of the SDF**

$$\mathcal{M}_{t,t+1}(\underline{w_{t+1}}) = \exp\left\{-r_t(\underline{w_t},\tilde{\theta}) + \alpha_t(\underline{w_t})' w_{t+1} - \psi_t[\alpha_t(w_t) \,|\, \underline{w_t};\theta_{\mathbb{P}}]\right\}.$$

  Once r_t has been specified $(r_t(\underline{w_t},\tilde{\theta}))$, $\alpha_t(\underline{w_t})$ remains to be specified. We assume that $\alpha_t(\underline{w_t})$ belongs to a parametric family:

$$\left\{\alpha_t(\underline{w_t};\theta_\alpha), \theta_\alpha \in \Theta_\alpha\right\}.$$

$$\begin{aligned} & \mathcal{M}_{t,t+1}(\underline{w_{t+1}},\theta) \\ = \quad & \exp\left\{-r_t(\underline{w_t},\tilde{\theta}) + \alpha_t(\underline{w_t},\theta_\alpha)' w_{t+1} - \psi_t\left[\alpha_t(\underline{w_t},\theta_\alpha)|\underline{w_t};\theta_{\mathbb{P}}\right]\right\}, \end{aligned} \tag{II.24}$$

  where $\theta = \left(\tilde{\theta}', \theta'_{\mathbb{P}}, \theta'_\alpha\right)' \in \tilde{\Theta} \times \Theta_{\mathbb{P}} \times \Theta_\alpha = \Theta$; $\tilde{\Theta}$ may be reduced to one point [Case i) of Remark II.3].

1) The Direct Modelling (cont'd)

- **Step 3: Internal consistency conditions (ICC)**
  For any payoff $g(\underline{w_s})$ delivered at $s > t$, with price $p(\underline{w_t})$ at t which is a known function of $\underline{w_t}$, we must have:

$$p_t[g(\underline{w_s})] = \mathbb{E}\left\{\mathcal{M}_{t,t+1}(\theta)\dots\mathcal{M}_{s-1,s}(\theta)g(\underline{w_s})|\underline{w_t}, \theta_{\mathbb{P}}\right\} \forall\ \underline{w_t}, \theta.$$

  These ICC pricing conditions may imply strong constraints on θ, for instance. When a component of w_t is a stock return, we have:

$$w_{1,t} = \log\frac{p_{1,t}}{p_{1,t-1}},\ \text{then } \mathbb{E}_t[\mathcal{M}_{t,t+1}\exp(e_1' w_{t+1})] = 1.$$

  If this component is a given yield:

$$w_{1,t} = -\frac{1}{h}\log B(t,h),\ \text{then } e_1' w_t = -\frac{1}{h}\log\mathbb{E}_t(\mathcal{M}_{t,t+h}).$$

- The previous three steps imply the specification of the R.N. dynamics:

$$\psi_t^{\mathbb{Q}}(u|\underline{w_t}, \theta_{\mathbb{P}}, \theta_\alpha) = \psi_t\left[u + \alpha_t(\underline{w_t}, \theta_\alpha)|\underline{w_t}, \theta_{\mathbb{P}}\right] - \psi_t\left[\alpha_t(\underline{w_t}, \theta_\alpha)|\underline{w_t}, \theta_{\mathbb{P}}\right].$$

- Here, (w_t) is chosen to be $\mathbb{P}$-affine, and the $\mathbb{Q}$-dynamics is a by-product (it is not chosen ex ante). If the $\mathbb{P}$-dynamics is Gaussian VAR, and if $r_t(\underline{w_t}, \tilde{\theta})$ and $\alpha_t(\underline{w_t}, \theta_\alpha)$ are affine functions of w_t, then the $\mathbb{Q}$-dynamics remains Gaussian VAR (with different parameters).

2) The R.N. constrained direct modelling (or mixed modelling)

- **Step 1: Specification of $\mathbb{P}$-dynamics.**
  We select a family $\{\psi_t(u|\underline{w_t}, \theta_{\mathbb{P}}), \theta_{\mathbb{P}} \in \Theta_{\mathbb{P}}\}$ to explain, for instance, some empirical stylized fact about w_t.
- **Step 2: Specification of $\mathbb{Q}$-dynamics.**
  We select a family $\{\psi_t^{\mathbb{Q}}(u|\underline{w_t}, \theta_{\mathbb{Q}}), \theta_{\mathbb{Q}} \in \Theta_{\mathbb{Q}}\}$ of affine $\mathbb{Q}$-dynamics, and possibly $\{r_t(\underline{w_t}, \tilde{\theta}), \tilde{\theta} \in \tilde{\Theta}\}$, to have explicit or quasi explicit pricing formula.

- We may want to choose $\mathbb{P}$- and $\mathbb{Q}$-dynamics such that, for any pair $(\psi_t, \psi_t^{\mathbb{Q}})$ belonging to these families, there exists a unique function $\alpha_t(\underline{w_t}, \theta_{\mathbb{P}}, \theta_{\mathbb{Q}})$ characterizing the exponential-affine SDF such that:

$$\psi_t^{\mathbb{Q}}(u|\underline{w_t}, \theta_{\mathbb{Q}}) = \psi_t\left[u + \alpha_t(\underline{w_t}, \theta_{\mathbb{P}}, \theta_{\mathbb{Q}})|\underline{w_t}, \theta_{\mathbb{P}}\right] - \psi_t\left[\alpha_t(\underline{w_t}, \theta_{\mathbb{P}}, \theta_{\mathbb{Q}})|\underline{w_t}, \theta_{\mathbb{P}}\right]. \quad (\text{II}.25)$$

  This condition is generally satisfied only for a subset of pairs $(\theta_{\mathbb{P}}, \theta_{\mathbb{Q}})$. The SDF is anyway a by-product.

- **Step 3: Internal Consistency Conditions (ICC).**
  Once the (constrained) parameterization $(\tilde{\theta}, \theta_{\mathbb{P}}, \theta_{\mathbb{Q}}) \in \tilde{\Theta} \times \Theta_{\mathbb{P}} \times \Theta_{\mathbb{Q}}$ is formalized, internal consistency conditions may be imposed.

- If the unique function $\alpha_t(\underline{w_t})$ that satisfies (II.25) does not exist, we can still adopt this mixed modeling strategy but in that case, the SDF is not exponential-affine.

3) The Back Modelling

- **Step 1: Specification of the R.N. dynamics** [and possibly of $\{r_t(\underline{w_t}, \tilde{\theta}), \tilde{\theta} \in \tilde{\Theta}\}$]: $\left\{\psi_t^{\mathbb{Q}}(u|\underline{w_t}; \theta_{\mathbb{Q}}), \theta_{\mathbb{Q}} \in \Theta_{\mathbb{Q}}\right\}$.
- **Step 2: Internal consistency conditions (ICC)**, if relevant (and $\forall\, \underline{w_t}, \tilde{\theta}, \theta_{\mathbb{Q}}$):

$$p_t[g(\underline{w_s})] = \mathbb{E}_t^{\mathbb{Q}}\left[\exp(-r_t(\underline{w_t}, \tilde{\theta}) - \cdots - r_{s-1}(\underline{w_s}, \tilde{\theta}))g(\underline{w_s})|\underline{w_t}, \theta_{\mathbb{Q}}\right].$$

  If $w_{1,t} = \log(p_{1,t}/p_{1,t-1})$, its ICC is $r_t = \psi_t^{\mathbb{Q}}(e_1 \mid \underline{w_t}; \theta_{\mathbb{Q}})$.
- **Step 3: Choice of the specification of** $\alpha_t(\underline{w_t})$ (without any constraint), providing the family $\{\alpha_t(\underline{w_t}, \theta_\alpha), \theta_\alpha \in \Theta_\alpha\}$. The $\mathbb{P}$-dynamics is a by-product:

$$\psi_t(u|\underline{w_t}; \theta_{\mathbb{Q}}, \theta_\alpha) = \psi_t^{\mathbb{Q}}\left[u - \alpha_t(\underline{w_t}, \theta_\alpha)|\underline{w_t}; \theta_{\mathbb{Q}}\right] - \psi_t^{\mathbb{Q}}\left[-\alpha_t(\underline{w_t}, \theta_\alpha)|\underline{w_t}, \theta_{\mathbb{Q}}\right].$$

- If the R.N. conditional p.d.f. $f_t^{\mathbb{Q}}(w_{t+1}|\underline{w}_t, \theta_1^*)$ is known in (quasi) closed form, the same is true for the historical conditional p.d.f.:

$$\begin{aligned} f_t(w_{t+1}|\underline{w}_t, \theta_{\mathbb{Q}}, \theta_\alpha) \quad = \quad & f_t^{\mathbb{Q}}(w_{t+1}|\underline{w}_t, \theta_{\mathbb{Q}}) \exp\left\{-\alpha_t'(\underline{w}_t, \theta_\alpha) w_{t+1} - \right. \\ & \left. \psi_t^{\mathbb{Q}}\left[-\alpha_t(\underline{w}_t, \theta_\alpha)|\underline{w}_t, \theta_{\mathbb{Q}}\right]\right\}, \end{aligned}$$

- Here, (w_t) is chosen to be $\mathbb{Q}$-affine, and the $\mathbb{P}$-dynamics is a by-product (it is not chosen ex ante).